Does the Gender Ratio at Colleges Affect High School Students' College Choices?*

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April 11, 2026

Abstract

Why do fewer female students enter STEM despite negligible gender gaps in mathematics and science? Using an incentivized discrete choice experiment with high school students, we show that both females and males prefer gender-balanced college programs and prefer being a majority to a minority, driven by anticipated difficulty fitting in. High-math-ability females show stronger minority avoidance in STEM, while males show the opposite. Combined with STEM's low female share, these preferences lead to inefficient talent allocation: high-math-ability students, especially females, forgo STEM, and low-math-ability students enter. These findings suggest STEM's gender composition itself deters female entry.

JEL Classification: J16, J24, I24

Keywords: STEM Gender Gap, College Choice, Gender Composition, Discrete Choice Experiment

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We thank Michihito Ando, Sonia Bhalotra, Maria Bigoni, Lukas Bolte, Jochem de Bresser, Caterina Calsamiglia, Damian Clarke, Lucia Corno, Dean Hyslop, Sota Ichiba, Clément Imbert, Shoya Ishimaru, Daiji Kawaguchi, Daiki Kishishita, Annalisa Loviglio, Akira Matsushita, Francesca Miserocchi, Takeshi Murooka, Johannes Rincke, Anna Salomons, Daan van Soest, Teodora Tsankova, Melanie Wasserman, Dajana Xhani, Kiyotaka Yageta, Atsushi Yamagishi, Basit Zafar, and participants at EEA-ESEM Rotterdam, the Trans Pacific Labor Seminar, Summer Workshop on Economic Theory, and seminars at Comenius University, Kobe University, Maastricht University, Tilburg University, the University of Florence, the University of Gothenburg, and the University of Osaka for their valuable feedback. We are deeply grateful to NPO Waffle for their valuable insights and support at the early stages of this project. Keigo Furukawa, Mifuyu Kira, Yoshino Kodama, and Keita Yamada provided excellent research assistance. We gratefully acknowledge financial support from JSPS KAKENHI (24K22636) and the Suntory Foundation (Torii Fellowship). The experiment was approved by the Institutional Review Board of the Tilburg School of Economics and Management (approval no. IRB FUL 2023-009) and preregistered on the AEA Registry under AEARCTR-0012577 (Inoue, Saito, and Takahashi 2023).

1 Introduction

Although the gender differences in mathematics and science are negligible at age 15 (OECD 2019), large gender gaps persist in Science, Technology, Engineering, and Mathematics (STEM) major choices at the college level: female students are 30.8 percentage points less likely than male students to major in STEM.[1] This gap may lead to inefficient talent allocation, as well as gender-biased research topics (Truffa and Wong 2025) and product innovation (Einiö, Feng, and Jaravel 2025; Koning, Samila, and Ferguson 2021).

One underexplored factor is that STEM programs are male-dominated, making female students a minority.[2] Being a minority carries well-documented costs, including a worse classroom environment (Gong, Lu, and Song 2021), less influence in study groups (Karpowitz et al. 2024), limited peer networking opportunities (Hampole, Truffa, and Wong 2024), and a higher likelihood of dropping out (Bostwick and Weinberg 2022). If students anticipate these costs when choosing a college program, they may avoid entering STEM altogether – potentially making gender composition self-perpetuating. Yet whether this is the case, and whether such anticipation generates inefficiency, remain open questions.

This paper studies whether the gender ratio at colleges affects high school students' college choices and whether the low female share in STEM leads to inefficient talent allocation. Answering these questions with observational data poses two challenges: the gender ratio correlates with other program attributes, making it difficult to isolate its effect; and most STEM programs are male-dominated, leaving little variation to identify how higher female shares would affect choices.

We address these challenges using an incentivized discrete choice experiment with high school students. The experiment elicits preferences over independently varied program attributes – including the student gender ratio across its full range – thereby isolating the effect of gender composition from confounding attributes. We incentivize truthful responses using the incentivized resume rating method (Kessler, Low, and Sullivan 2019), providing students with career advice tailored to their choices. We validate experimental choices against actual choices several months later.

We find that the gender ratio affects both female and male students' college choices. Both prefer gender-balanced programs over male- or female-majority programs, and both prefer being a majority to being a minority. Relative to a gender-balanced program (50% own gender), being a minority (25% own gender) reduces choice probability by 11 percentage points for female students and 9 percentage points for male students; being a majority (75% own gender) reduces choice probability by 6.5 percentage points for both genders. A decomposition reveals that students avoid being a minority primarily because they anticipate difficulty fitting in.

We also find that preferences for the gender ratio differ between STEM and non-STEM programs in ways that matter for talent allocation. Female students in the top half of mathematics ability

1. Based on the number of bachelor program graduates in 2023 from OECD Data Explorer, "Number of enrolled students, graduates and new entrants by field of education" dataset (accessed November 6, 2025).

2. A notable exception is Ersoy and Speer (2025), who show that providing non-job-related major information to students, including student gender composition, changes their major choices.

exhibit stronger minority avoidance in STEM than in non-STEM: relative to a gender-balanced program, being a minority (25% female) reduces their choice probability by 14 percentage points in STEM versus 7 percentage points in non-STEM – a twofold difference. Male students show the opposite pattern: they are less sensitive to gender composition in STEM than in non-STEM. Being a minority (25% male) reduces male students' choice probability by 8 percentage points in STEM versus 12 percentage points in non-STEM. These preferences imply a potential inefficiency in talent allocation.

We then quantify the inefficiency in talent allocation by comparing the baseline scenario – where the female share in STEM is 25% – against a benchmark scenario of the best allocation attainable where students' preferences for the gender ratio are removed.[3] In our exercise, the low female share in STEM reduces the average STEM student's mathematics ability by 0.2 standard deviations. This drop arises because students with high mathematics ability – especially female students – are deterred from entering STEM, and their seats are filled by students with low mathematics ability. The average non-STEM student's reading ability rises, but by only one-third as much.

Finally, we examine whether students choose their preferred field, using actual choices collected several months after the experiment. Most students (77.1%) choose according to their preferences, but female students who prefer STEM are less likely to follow through than their male counterparts. Since our classification captures intrinsic field preferences but not preferences over gender composition, this gap is consistent with our main finding: female students who prefer STEM may ultimately avoid it due to anticipated difficulty fitting into a male-dominated environment.

Related Literature Our paper contributes to several strands of literature. First, we identify a novel source of the STEM gender gap in higher education: the gender composition itself. Prior work has identified numerous contributing factors, which we group into three categories. (i) *Individual factors* include male students' stronger preferences for competition (Buser, Niederle, and Oosterbeek 2014) and female students' comparative advantage in non-STEM (Breda and Napp 2019; Goulas, Griselda, and Megalokonomou 2024). (ii) *Environmental factors* include gender-biased teachers (Carlana 2019; Miserocchi 2024), parental expectations (Carlana and Corno 2024), lack of role models (Breda et al. 2023; Carrell, Page, and West 2010; Riise, Willage, and Willén 2022; Riley 2024), non-cooperative classroom dynamics (Di Tommaso et al. 2024), insufficient emphasis on STEM's social relevance (Long and Takahashi 2025), and peer effects (Bechichi and Kenedi 2024; Valdebenito 2023; Mouganie and Wang 2020; Fischer 2017).[4] (iii) *Anticipated returns* include preferences for job amenities (Burbano, Padilla, and Meier 2024; Zafar 2013; Wiswall and Zafar 2018) and marriage-market considerations (Wiswall and Zafar 2021). Our effect size – an 11-percentage-point increase in female students' choice probability when moving from 25% to 50% female – is economically meaningful: it corresponds to roughly one-fourth of the baseline STEM

3. Setting the STEM female share to 50% instead of removing gender preferences yields essentially the same results.

4. These studies show that peers' and recent graduates' field choices and academic performance affect students' own field choices, with effects often stronger among female students.

choice probability among female students under no constraints.[5]

Within this literature, our study is closest to Ersoy and Speer (2025), who show that providing gender composition information changes students' major choices. We go further by recovering preferences over the full range of female shares and identifying the mechanisms underlying these preferences. Our paper also relates to Carlana and Corno (2025), who show that female junior high students are less likely to choose counter-stereotypical tasks when they expect male-dominated peer groups. We extend this work by eliciting preferences over actual college attributes, and our private response design rules out social image concerns as a driver.

Second, we extend the emerging literature on preferences for gender diversity from workplace to educational settings. Existing work shows that workers – especially women – prefer gender-diverse workplaces, and that this preference contributes to occupational segregation (Funk, Iriberri, and Savio 2024; Högn et al. 2025; Schuh 2024). We document analogous preferences among high school students' college choices, suggesting these preferences form before labor market entry. This implies that interventions targeting educational choices may be effective in facilitating STEM-related human capital accumulation among women. Our finding that making STEM programs more gender-balanced improves the allocation of talent aligns with documented benefits of gender diversity,[6] including reduced harassment risk (Folke and Rickne 2022), better recognition of ideas and talent (Koffi 2025; Bello, Casarico, and Nozza 2025), and improved team performance (Hoogendoorn, Oosterbeek, and van Praag 2013).[7]

Finally, our findings inform policies aimed at closing the STEM gender gap. While such policies – role model exposure (Breda et al. 2023; Riise, Willage, and Willén 2022; Riley 2024),[8] mentorship (Canaan and Mouganie 2023; Carrell, Page, and West 2010), single-sex instruction (Booth, Cardona-Sosa, and Nolen 2018), and pedagogical changes (Di Tommaso et al. 2024; Long and Takahashi 2025)[9] – are often justified on equity grounds, our results provide a complementary efficiency-based rationale.[10] This may help address concerns from those who view such policies as zero-sum.[11]

The remainder of the paper is structured as follows. Section 2 explains the institutional background. Section 3 details the experimental design. Section 4 describes the summary statistics of the experimental data. Section 5 presents the main results. Section 6 quantifies the degree of inefficient allocation of talent. Section 7 validates experimental choices against actual choices.

5. The baseline STEM choice probability among female students in our sample is approximately 42.1% under no constraints.

6. On the negative side, Shan (2024) finds that students adopt more traditional gender roles when assigned to a mixed-gender study group in college.

7. In the context of family socioeconomic status (SES), Tadjfar and Vira (2025) find that students from low-SES backgrounds are reluctant to apply for elite colleges dominated by students from high-SES backgrounds due to concerns about fitting in.

8. Porter and Serra (2020) show the effectiveness of female role models in inspiring female college students to pursue an economics major.

9. Avery et al. (2024) and Owen and Hagstrom (2021) also find similar results in economics curricula.

10. While not in education, Baltrunaite et al. (2014) and Besley et al. (2017) show that a gender quota for political candidates improved elected politicians' competence. We expect similar effects for women in STEM in higher education.

11. For example, The Economist (2024) shows that some young men in Europe consider that these policies may reduce their opportunities.

Section 8 concludes.

2 Institutional Background

We conduct our experiment in the Greater Tokyo area in Japan, which offers an ideal setting for several reasons. First, the pathway from high school to college follows a relatively uniform institutional structure compared with Europe or the US, allowing us to define hypothetical programs using attributes that students commonly observe and understand. Second, colleges and programs are ranked by a widely recognized selectivity index, which allows us to control for perceived program prestige and expected labor-market prospects. Third, college programs are densely concentrated in the Greater Tokyo area, making the hypothetical program attributes appear realistic to students and limiting unstated geographic and financial constraints on their choices.

High School High school runs from grades 10 to 12, typically from age 15 to 18 in Japan. Though not compulsory, nearly 99% of junior high school graduates attend high school (Ministry of Education, Culture, Sports, Science and Technology 2021). College enrollment rates are also high, at nearly 60%.[12] However, colleges vary widely in academic orientation: while some are research-focused, many function more like vocational institutions. We refer to the former as selective colleges and the latter as non-selective colleges. Because STEM fields require strong mathematical preparation, research-oriented STEM programs are concentrated at selective colleges. As such, academic high schools are the primary pathway for students aiming to enter selective colleges or research-oriented STEM programs.

Students at academic high schools choose a track at the end of grade 10, which determines the subjects they study in grades 11 and 12. There are two tracks: humanities and sciences. In the humanities track, students study advanced reading (advanced Japanese), English, history/social studies, and mathematics. In the sciences track, students study reading (Japanese), English, sciences (biology, chemistry, and/or physics), and advanced mathematics. A similar track system exists in many European countries.[13]

Students in these academic high schools regularly take mock exams to prepare for college entrance exams. On the score reports for these mock exams, students can see the likelihood of admission to their preferred programs, calculated based on their exam scores and the program's selectivity, which they use to make and adjust their study plan. To help students select programs, nearly all academic high schools maintain counselor offices with information about each program, including selectivity, gender ratio, and other attributes used in this paper's experiment.

12. As of 2023: https://www.ipss.go.jp/syoushika/tohkei/Data/Popular2024/T11-03.htm (accessed June 9, 2025).

13. In Italy, students choose their track between humanities and sciences at the end of grade 9 (see, for example, Carlana and Corno 2025). In France, students choose their track between humanities, social sciences, and sciences at the end of grade 10 (see, for example, Breda et al. 2023). In the Netherlands, students in academic secondary schools (VWO) choose their track between science, health, social sciences, and humanities at the end of grade 9 (see, for example, Buser, Niederle, and Oosterbeek 2014).

College Application As in Europe but unlike in the US (Bordon and Fu 2015), students apply to specific college programs and cannot change their majors later. Programs differ in their attributes, such as major, selectivity, tuition, whether they are public or private, and location, among others. Since living alone can be costly and some parents prefer their children not to live alone, many college students live with their parents. Among various locations in Japan, the Greater Tokyo area offers the greatest variety of college attributes: nearly 29% of all colleges in Japan are located there, where about 41% of all college students study.[14]

Most selective college programs employ an exam-based, meritocratic admissions system. They rank applicants by exam score and make offers from the top of the list. However, each program requires exams in different subjects. Humanities and social sciences programs usually require exams in advanced reading, English, history/social studies, and mathematics. On the other hand, science, engineering, and medicine programs typically require exams in advanced mathematics, English, sciences, and reading. Since students who choose the science track in high school do not study advanced reading and history/social studies, they are effectively constrained from applying to most humanities and social sciences programs. Similarly, students who choose the humanities track in high school do not study advanced mathematics and sciences, making it difficult to apply to science, engineering, and medicine programs. In this way, high school track choice largely determines the set of college majors students can realistically pursue.

Each college program is assigned a single number called the "selectivity index" by commercial college entrance exam preparation companies. The index represents the difficulty of admission and the selectivity (or prestige) of the program and is calculated based on admitted students' mock exam performance. The index is expressed as a z-score rescaled to have a mean of 50 and a standard deviation of 10. The selectivity of the program from which a student graduates significantly affects the quality of the first job they obtain (Nakajima 2018) as well as promotions in the first few years after starting that job (Araki, Kawaguchi, and Onozuka 2016). Given the rigid labor market and limited job mobility in Japan (Moriguchi 2014), the quality of the first job affects students' career prospects more than in other OECD countries (Genda, Kondo, and Ohta 2010).[15]

3 Experimental Design

To investigate whether the gender ratios at colleges affect high school students' college choices, we conducted an incentivized discrete choice experiment at four selective academic high schools in the Greater Tokyo area in Japan. We integrated this experiment as a "career planning module" within the 10th-grade curriculum of the participating high schools from December 2023 to July 2024. The experiment was conducted in person at three high schools and asynchronously online at one high school.

14. From an article by an educational materials and news publishing company Obunsha: https://eic.obunsha.co.jp/file/educational_info/2024/1022.pdf (accessed July 31, 2025).

15. Genda, Kondo, and Ohta (2010) find that those who entered the labor market during a recession suffer worse employment conditions in terms of unemployment and earnings in Japan than in the US.

The experiment lasted about 40 minutes on average, including distributing the participation gifts. A total of 628 students took part, with 619 providing valid responses (311 female, 298 male, 10 non-binary). Since this study focuses on binary gender, we excluded responses from non-binary students, resulting in 609 responses with 15 choices each, for a total of 9135 choices.

3.1 Sample Selection

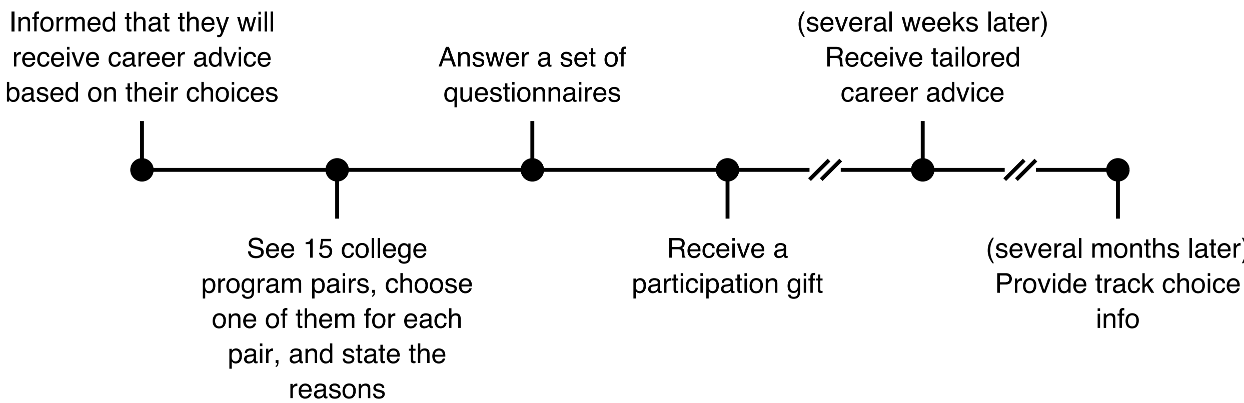
Schools We contacted teachers at academic high schools in the Greater Tokyo area through our network and obtained their consent to conduct the experiment as part of their school curriculum within a career guidance class. This was intended to make the experiment feel natural to students and to ensure dedicated time for participation. We restricted our potential sample to academic high schools in the Greater Tokyo area for three reasons. First, we wanted to ensure the experimental content was relevant to students: the experiment focused on college choices, and students needed to be planning to attend college. Based on the schools' placement records, over 95% of the recent graduates attended college. Second, we wanted to include students whose mathematics skills did not constrain their major choices: as discussed in Section 2, most STEM programs require good mathematics abilities, and students from academic high schools are the primary group prepared for STEM. Third, we wanted to prevent students from implicitly considering potential location and financial constraints when making their choices: as discussed in Section 2, the Greater Tokyo area offers the widest variety of college programs, making such constraints less relevant there. We also wanted the attributes of the hypothetical college programs to appear natural to students, and the Greater Tokyo area was suitable for this purpose as well.

Students The teachers at the participating high schools distributed the information letter and consent form to guardians of all 10th-grade students, except at one school where only one class participated. We restricted the sample to 10th-grade students before their track choice because the track choice restricted the college programs they could apply for, as discussed in Section 2. The information letter did not mention that the experiment was about gender ratios or STEM to minimize the experimenter demand effect (Zizzo 2010). Instead, we explained that students would evaluate 15 hypothetical college programs, answer a short questionnaire, and receive a tailored career advice sheet based on their evaluations. We also clarified that the data from the experiment would be used for academic research to improve education policy. Nearly 90% of the guardians and students provided consent and participated.

3.2 Flow of the Experiment

Students were first told that they would receive a career advice sheet based on their choices in the experiment. We promised them that their responses and the career advice sheet would not be shared with anyone, including guardians, teachers, or peers, to minimize their potential influence, as previous studies suggest that they can affect students' study choices (Carlana 2019; Carlana and Corno 2024; Giustinelli 2016; Miserocchi 2024; Müller 2024; Valdebenito 2023). The students then

Figure 1: Flow of the Experiment



Notes: This figure shows the flow of the experiment.

saw 15 program pairs one by one and chose the one they would prefer to attend; see Figure 2 for an example of a hypothetical program pair. Additionally, for each pair, they indicated a program that applied better to each of four statements. The statements were developed prior to the experiment, based on an open-ended questionnaire in a pre-test conducted with a group of students different from the participants in this experiment about why they care about the gender ratios.

After making choices for the 15 program pairs, students completed a questionnaire about their demographics, academic abilities, behavioral traits, and beliefs about the gender ratios in average college programs in Japan. Appendix Section C presents the questionnaire. Afterward, students received a participation gift (a set of decorative, functional pens) equivalent to 500 JPY (approx. 5.27 USD in 2022 PPP) for their participation.[16]

Several weeks later, each student received a career advice sheet we created. The sheet had two parts: a tailored part and a non-tailored part. The tailored part included the top three attributes that students cared about most, along with the top reason they prioritized when choosing a college program, based on their choices in the hypothetical programs. The non-tailored part contained non-individualized information useful for most high school students, such as tips for choosing college programs, college admissions, financing, studying abroad, attending graduate school, and finding jobs.

Finally, several months after the experiment, we requested students' track choice data from schools. Some schools provided data directly from their records; others provided it through a follow-up survey completed by students.[17] In total, we obtained valid responses from 461 out of 609 students (or 75.7% of students).

16. USD to JPY PPP was 94.93 in 2022: https://www.oecd.org/en/data/indicators/purchasing-power-parities-ppp.html (accessed November 18, 2024).

17. Approximately 8-15 months. It is 15 months because one school had their track choice at the end of the 11th grade.

Figure 2: Hypothetical Program Pair

Pair 4/15

AB College Dept. of Literature		AX College Dept. of Engineering
	Dept. Characteristics	
57.5	Selectivity index	62.5
700	Cohort size	600
35% male, 65% female	Student gender ratio	55% male, 45% female
	College Characteristics	
Yes	Intl exchange program	No
65%	Club participation rate	45%

Which program would you like to attend?

AB College, Dept. of Literature

AX College, Dept. of Engineering

Which program do you feel these statements apply to more?

	AB College, Dept. of Literature	AX College, Dept. of Engineering
I can do well in my studies	○	○
I can find a career I want to pursue	○	○
I can fit in	○	○
I can meet inspiring seniors	○	○

Notes: This figure shows an example of a hypothetical program pair students would see during the experiment.

3.3 Attributes

We randomly assigned attributes to each program, including college name, department, department selectivity index, department cohort size, department student gender ratio, whether the college has an international exchange program, and college club activity participation rate. These are attributes the students can find in college program guidebooks available at the school counselor's room, which they can read at any time.[18]

Our main attributes of interests are (i) student gender ratio and (ii) department, which indicates whether the program is STEM or non-STEM. We included other attributes to make the programs appear more realistic to students and selected attribute value ranges that are plausible for students in our sample to reduce hypothetical bias (List and Shogren 1998; List, Sinha, and Taylor 2006). We asked them to assume that attributes not shown were identical between the programs.

College names consist of two uppercase letters and we draw them without replacement for each program in a pair from a list ranging from AA to BD to make sure they were unrelated to the actual college names. The department was drawn from a list of 12 popular departments among college students, of which 6 were STEM and 6 were non-STEM.[19] First, we randomly assigned either STEM or non-STEM to one program in the pair. If STEM was selected, then the other program was assigned non-STEM with 75% probability and STEM with 25% probability to reduce the chances that both programs are in the same category. Next, we selected a specific department within the relevant list. STEM departments included Physics, Chemistry, Biology, Engineering, Information Technology, and Agriculture. Non-STEM departments included Literature, Law, Business, Economics, Sociology, and Foreign Language. We excluded Medicine and Education, as both are popular but lead to specialized careers such as doctors, nurses, pharmacists, and teachers, which differ from most student career paths, and the attributes used may not be very relevant for these programs.

Other attributes include selectivity index, which ranges from 55 to 72.5 with an increment of 2.5; cohort size, which varies from 200 to 900 with an increment of 50; student gender ratio, which spans from 5% to 95% for females and males but sums to 100%; whether the college has an international exchange program, which has an 80% chance of being "Yes" and a 20% chance of being "No"; and club participation rate, which ranges from 40% to 85% with an increment of 5%.[20] These attribute values were drawn with replacement for each program. Appendix Table B3 shows the possible values for each attribute.

We used a hypothetical choice experiment because it allowed us to elicit students' preferences

18. Some of these guidebooks provide club participation as qualitative information. Its range is obtained from the National Federation of University Cooperative Associations (2023).

19. We first referred to the Ministry of Education, Culture, Sports, Science and Technology (MEXT) department-classification list (https://www.mext.go.jp/component/b_menu/other/__icsFiles/afieldfile/2018/08/02/1407357_4.pdf, accessed June 30, 2023) to compile a comprehensive set of programs. We then consulted a college-choice guide for high school students (https://shingakunet.com/journal/column/20210415000002/, February 21, 2023 version) and selected 12 popular programs.

20. We set the probability that a program has an international exchange program to 80% because most colleges in Japan have one.

over attributes that were varied independently. This design was crucial for three reasons: most STEM programs are male-dominated, leaving little variation in gender ratios; the gender ratio correlates with other program attributes; and using actual college names would lead students to infer attributes not shown.

3.4 Incentives

One concern with hypothetical choice experiments is that students may lack incentives to state their true preferences without real consequences. Although Hainmueller, Hangartner, and Yamamoto (2015) show that choices in hypothetical vignettes and actual behaviors are highly correlated, we address this concern using the incentivized resume rating method (Kessler, Low, and Sullivan 2019), which involves providing career advice based on students' choices.[21] Assuming students believed that we researchers held potentially valuable new information, the method is incentive compatible: the expected value of the advice increases with the truthfulness of their choices. Because students were from academic high schools and planned to attend selective colleges, we assume most believed the researchers had valuable academic and career information.[22]

Specifically, we provided the following information in the information letter and at the beginning of the experimental instructions. This wording closely followed the original incentivized resume rating studies (Kessler, Low, and Sullivan 2019; Low 2024):

> *Through this module we will give you information relevant for your career choice. You will complete it on the internet using a laptop or a tablet. It is expected to last for 35 minutes and consists of:*
>
> - *Evaluation of 15 hypothetical program pairs*
> - *A short questionnaire*
>
> *We will send you a career advice sheet created based on your evaluation.*

4 Data

4.1 Variable Construction

Academic Abilities We convert students' academic abilities, obtained through a post-experimental questionnaire, into population z-scores to make them comparable across schools and interpretable within the entire student pool. To do this, we use the latest placement records of graduates from

21. Several studies employed the incentivized resume rating method to elicit preferences for attributes that are hard to obtain from revealed preferences. Low (2024) elicited heterosexual adults' preferences for dating partners by providing dating advice from a dating coach based on their ratings of hypothetical opposite-gender partner profiles. Macchi (2023) elicited loan officers' preferences for borrowers by offering referrals to loan clients based on their ratings of hypothetical borrower profiles. Gallen and Wasserman (2023) elicited college students' mentor preferences by providing mentor characteristics that students care most about. Chan (2024) elicited patients' preferences for doctors by offering booking options based on patients' choices.

22. At the time of the experiment, Inoue obtained a PhD from one of Japan's most prestigious colleges, Saito earned a master's degree in the US and has rich industry experience, and Takahashi earned a PhD from a European university.

each participating high school, assign the selectivity index to each college in the records, and rank the placements by the selectivity index. We then assign each student a selectivity index based on their academic rank within their high school in specific subjects (reading, mathematics, English, and total). The selectivity index for each program is obtained from the list prepared by Kawaijuku in 2024, one of the most popular commercial college exam preparation companies in Japan.[23] Since public colleges require a larger number of subjects in the entrance exam, we add 0.5 to the index of public colleges (equivalent to 5 points in the raw index), following Araki, Kawaguchi, and Onozuka (2016).

Behavioral Traits We elicit students' behavioral traits through a post-experimental questionnaire: confidence in reading, mathematics, and English; competitiveness; and risk-taking, all rated on a 5-point Likert scale with 3 being neutral.[24] For confidence questions, we ask how accurately their recent exam scores, entered on the previous page, reflected their ability. We convert these 5-point scales to the range [-1, 1] for better interpretability, with 0 being neutral.

4.2 Summary Statistics

Table 1 presents summary statistics for 311 female and 298 male students in the final sample (totaling 609 students), along with their differences. Panel A presents students' demographics and indicates no differences in parents' education levels or parental investments (proxied by extra schooling days per week) between female and male students. Panel B presents students' academic abilities and shows that female students outperform male students in reading and English, while male students excel in mathematics; female students have a slight edge in overall scores.[25] As expected given our sample of academic high schools, students perform approximately 0.6 to 0.7 standard deviations above the national average. Panel C presents students' behavioral traits, showing that male students are more confident in their mathematics and English abilities – despite lower English performance – and are less risk-averse than female students, consistent with existing literature on gender differences in preferences (Croson and Gneezy 2009). Although statistically insignificant, the gender difference in competitiveness is also consistent with existing literature, with male students more competitive than female students.

5 Gender Ratio Affects Students' College Choices

5.1 Econometric Framework

Estimation of Preferences To estimate students' preferences for program attributes, we assume that student i of gender g's preferences over program d with attributes X in pair j are represented

23. https://www.keinet.ne.jp/exam/ranking/index.html (accessed December 18, 2024)
24. The questionnaire-based competitiveness measure is adapted from Buser, Niederle, and Oosterbeek (2024), and the risk-taking measure is adapted from Dohmen et al. (2011).
25. Appendix Figure B1 shows the distribution of abilities for female and male students.

Table 1: Summary Statistics of Students in the Final Sample

	Female (N=311)	Male (N=298)	Difference (M – F)
Panel A: Demographics			
Mother bachelor or above	0.59	0.56	-0.03
	(0.49)	(0.50)	(0.04)
Father bachelor or above	0.80	0.77	-0.03
	(0.40)	(0.42)	(0.04)
Both bachelor or above	0.50	0.48	-0.02
	(0.50)	(0.50)	(0.04)
Extra schooling (no. days/week)	0.94	0.92	-0.02
	(1.06)	(1.04)	(0.09)
Panel B: Academic abilities (population z-score)			
Reading score	0.82	0.59	-0.23***
	(0.74)	(0.75)	(0.06)
Math score	0.65	0.77	0.12**
	(0.75)	(0.75)	(0.06)
English score	0.82	0.58	-0.25***
	(0.69)	(0.78)	(0.06)
Total score	0.71	0.61	-0.11*
	(0.74)	(0.76)	(0.06)
Panel C: Behavioral traits			
Reading confidence [-1,1]	-0.01	0.05	0.06
	(0.39)	(0.49)	(0.04)
Math confidence [-1,1]	-0.02	0.18	0.20***
	(0.39)	(0.50)	(0.04)
English confidence [-1,1]	-0.00	0.10	0.10***
	(0.40)	(0.45)	(0.03)
Competitiveness [-1,1]	0.03	0.11	0.08
	(0.67)	(0.68)	(0.05)
Risk-taking [-1,1]	-0.36	-0.18	0.18***
	(0.61)	(0.70)	(0.05)

Notes: This table presents summary statistics of 311 female and 298 male students in the final sample (609 students in total) as well as their differences. Standard deviations and standard errors are in parentheses. Significance levels: * 10%, ** 5%, and *** 1%.

by a linear indirect utility function:

$$V_{ijd} = X'_{jd}\beta^g + \varepsilon_{ijd} \tag{1}$$

The probability that student i chooses the right program r over left l in choice pair j is then:

$$P(V_{ijr} > V_{ijl}|X, g) = F((X_{jr} - X_{jl})'\beta^g) \tag{2}$$

where F is the cumulative distribution function (CDF) of $\varepsilon_{ijr} - \varepsilon_{ijl}$. We assume an identity function for the CDF, $F(x) = I(x) = x$, and estimate the model using OLS for ease of interpretation and decomposition. The empirical model we estimate is therefore:

$$C_{ij}^r = \alpha^g + \zeta^g(FShare_{jr} - FShare_{jl}) + (W_{jr} - W_{jl})'\omega^g + \epsilon_{ij} \tag{3}$$

where C_{ij}^r is an indicator variable equal to 1 if student i chooses the right program in choice pair j, $FShare_{jd}$ is the share of female students in program d within pair j, $W_{jd} \equiv X_{jd} \setminus \{FShare_{jd}\}$ is a vector of program d's attributes in pair j excluding the female student share, and α^g is the intercept for the right program.

We also present the results with logit (assume $F(x) = \Lambda(x)$) as a robustness check.

Decomposition of the Choices To investigate the underlying reasons for students' program choices, we treat the four reasons we elicited in the experiment as mediators: fit in, role model, studies, and career. We then decompose the treatment effects of the female student share on the choices into these four reasons, following Gelbach (2016) and Gong, Lu, and Song (2021).

Replace C_{ij}^r in equation 3 with the four reasons:

$$M_{ij}^m = \kappa^{m,g} + \xi_{m,g}(FShare_{jr} - FShare_{jl}) + (W_{jr} - W_{jl})'\psi^{m,g} + \nu_{ij}^m \tag{4}$$

where M_{ij}^m (for $m = 1, 2, 3, 4$) is an indicator variable equal to 1 if student i indicated that reason m better applies to the right program in the choice pair j.

Finally, include all the M_{ij}^ms in equation 3:

$$C_{ij}^r = \alpha^{g,full} + \zeta^{g,full}(FShare_{jr} - FShare_{jl}) + (W_{jr} - W_{jl})'\omega^{g,full} + \sum_{m=1}^{4} \eta^{m,g} M_{ij}^m + \epsilon_{ij}^{full} \tag{5}$$

Gelbach (2016) shows that:

$$\hat{\zeta}^g = \hat{\zeta}^{g,full} + \sum_{m=1}^{4} \hat{\eta}^{m,g}\hat{\xi}^{m,g} \quad \forall g \tag{6}$$

where $\hat{\eta}^{m,g}\hat{\xi}^{m,g}$ is the part of the treatment effects $\hat{\zeta}^g$ explained by reason M_{ij}^m, and $\hat{\zeta}^{g,full}$ is the part of the treatment effects $\hat{\zeta}^g$ unexplained by any of the four reasons.

We discretize the female student share as discussed later, so we perform this decomposition for each bin of the share. We first present the results of the preference estimations (Section 5.2), then apply this decomposition to understand underlying reasons (Section 5.3).

Table 2: Preferences for Program Attributes

Sample:	Female		Male		All	
Outcome:	Choose the program (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
STEM	-0.079***	-0.079***	0.038**	0.038**	0.038**	0.038**
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)
Female student share	0.090***	0.821***	-0.036	0.818***	-0.036	0.818***
	(0.022)	(0.088)	(0.024)	(0.089)	(0.024)	(0.089)
Female student share squared		-0.745***		-0.875***		-0.875***
		(0.088)		(0.090)		(0.090)
Selectivity index (population SD)	0.046***	0.047***	0.075***	0.075***	0.075***	0.075***
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Cohort size/100	0.003	0.003	0.006**	0.006**	0.006**	0.006**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Intl exchange program	0.059***	0.060***	0.033**	0.034**	0.033**	0.034**
	(0.014)	(0.014)	(0.013)	(0.013)	(0.013)	(0.013)
Club participation rate	0.193***	0.196***	0.122***	0.130***	0.122***	0.130***
	(0.042)	(0.042)	(0.041)	(0.040)	(0.041)	(0.040)
Female					0.016	0.015
					(0.011)	(0.011)
STEM x Female					-0.117***	-0.116***
					(0.024)	(0.024)
Female student share x Female					0.126***	0.003
					(0.032)	(0.126)
Female student share squared x Female						0.131
						(0.126)
Selectivity index (population SD) x Female					-0.029*	-0.028*
					(0.016)	(0.016)
Cohort size/100 x Female					-0.003	-0.003
					(0.004)	(0.004)
Intl exchange program x Female					0.027	0.026
					(0.019)	(0.019)
Club participation rate x Female					0.071	0.066
					(0.058)	(0.058)
Constant	0.503***	0.503***	0.488***	0.487***	0.488***	0.487***
	(0.007)	(0.007)	(0.008)	(0.008)	(0.008)	(0.008)
Adj. R-squared	0.036	0.054	0.021	0.045	0.029	0.050
No. observations	4649	4649	4451	4451	9100	9100
No. students	310	310	297	297	607	607

Notes: This table presents the OLS coefficient estimates on the program attributes with choice as the dependent variable. Columns 1 and 2 present estimates for female students, columns 3 and 4 present estimates for male students, and columns 5 and 6 present estimates for differences between female and male students. Standard errors are clustered at the student level. Significance levels: * 10%, ** 5%, and *** 1%.

5.2 Preferences for the Gender Ratio

Table 2 presents preferences for program attributes.[26] First, female students are 7.9 percentage points less likely to choose STEM programs, while male students are 3.8 percentage points more likely to do so, consistent with the literature. Second, both female and male students prefer

26. Appendix Table B4 presents the same specifications but with indicator variables for the four reasons instead of choice as the outcome variables.

programs with higher selectivity indices, but male students show a slightly stronger preference: a one-standard-deviation increase in selectivity raises female students' choice probability by 4.7 percentage points and male students' by 7.5 percentage points – a 2.8 percentage points gender gap.

Third, students also favor the social aspects of the programs, consistent with the evidence that students value non-academic amenities in college (Ersoy and Speer 2025; Gong et al. 2021; Jacob, McCall, and Stange 2018). Both female and male students prefer programs that have an international exchange program: having an international exchange program increases female students' choice probability by 6.0 percentage points and male students' choice probability by 3.4 percentage points. Club participation rates also affect students' choices: a 10 percentage points increase in club participation rate increases female students' choice probability by 2.0 percentage points and male students' choice probability by 1.3 percentage points. The effect of the cohort size is quantitatively minimal: an increase of a program size by 100 students increases male students' choice probability by 0.6 percentage points but has no statistically significant effect on female students. Logit estimates in Appendix Table B5, where we convert the coefficient estimates into average marginal effects show essentially the same results as those with OLS.

The effect of female student share is highly non-linear: the coefficient on the squared term is significant for both genders (columns 2 and 4). We therefore discretize the female share into 19 bins and re-estimate with 50% as the baseline.

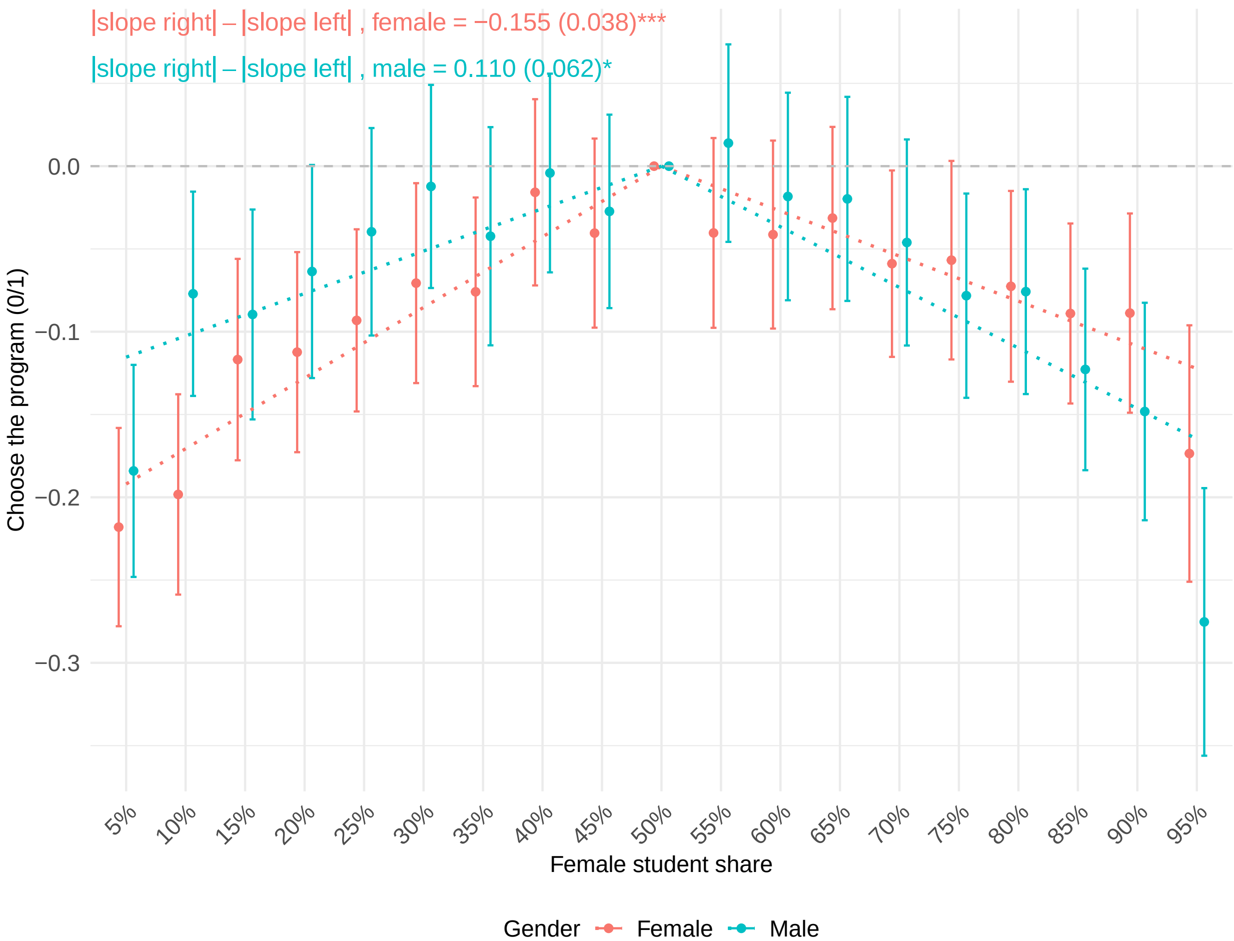
Figure 3 reveals three patterns. First, both genders prefer gender-balanced programs over those where they would be a minority: relative to 50% female, being a minority (25% own gender) reduces choice probability by 11 percentage points for female students and 9 percentage points for male students. Second, both genders also prefer balance over being a strong majority: being 75% own gender reduces choice probability by 6.5 percentage points for both. Third, despite this preference for balance, students still prefer majority to minority status – the fitted slopes are steeper on the minority side than the majority side, indicating stronger aversion to minority status than to majority status.

5.3 Underlying Reasons

Figure 4 decomposes the effect of gender composition into four stated reasons – fitting in, finding a role model, doing well in studies, and finding a career – plus an unexplained residual.

Panel A shows that the main reason the gender ratio affects female students' program choices varies significantly depending on whether most students in the program are male or female. When the majority of students are male, the gender ratio affects female students' program choices primarily due to their concerns about fitting in. On the other hand, when the majority of students are female, multiple factors – not only concerns about fitting in, but also concerns about finding a role model, doing well in studies, finding a career, and unexplained reasons – all contribute to how the gender ratio affects their choices. One notable observation is that the unexplained part is larger when the majority of students are female. It may reflect marriage market concerns, as students often find their future partners in the field of study within their college (Artmann et al. 2021; Kirkebøen

Figure 3: Preferences for the Gender Ratio



Notes: This figure plots the coefficient estimates and the 95% confidence intervals of female student share discretized into 19 equally spaced bins, with 50% as the baseline, separately for female students (red) and male students (blue). Standard errors are clustered at the student level. The dotted lines are weighted least squares linear fits of each point, with inverse variance weighting, for female and male students on both sides (below 50% and above 50%). We imposed the constraint that the lines pass through the 50% point.

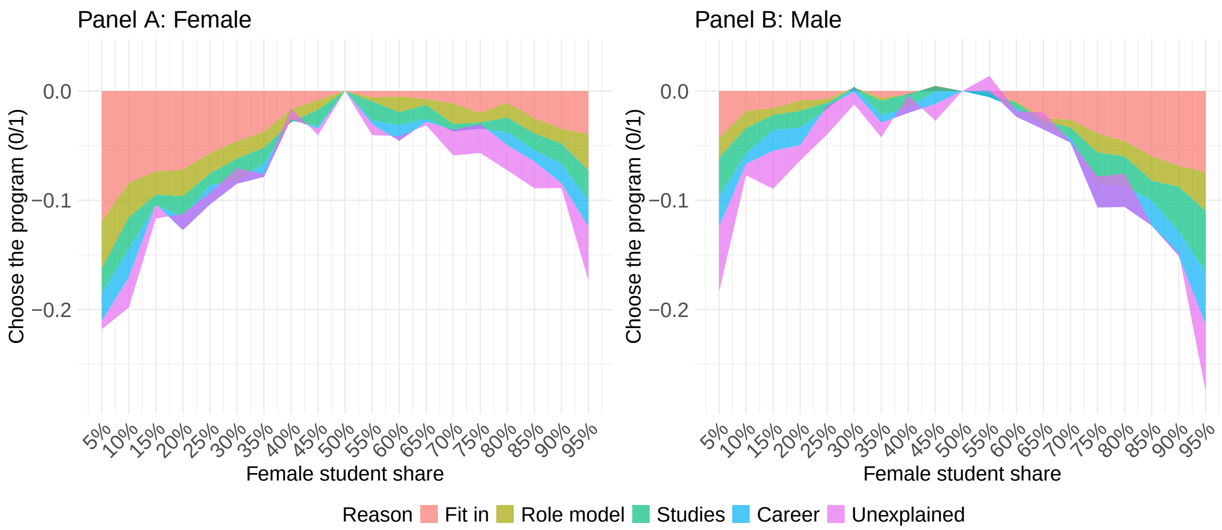
et al. 2025; Pestel 2021), especially in fields with a larger fraction of opposite-gender peers (Artmann et al. 2018).

Interestingly, Panel B shows that the patterns are very similar for male students: when most students are female, the gender ratio affects male students' program choices primarily through the expected difficulty in fitting in. When most students are male, however, all four reasons, along with other reasons beyond these four, explain their choices.

5.4 Heterogeneity of Preferences

Figure 5 plots coefficient estimates and the 95% confidence intervals for female and male students, where we interact the attributes with the STEM dummy, with 50% as the baseline for both programs. Panels A and B show all students, Panels C and D restrict the sample to students among the top

Figure 4: Decomposition of Preferences for the Gender Ratio



Notes: This figure plots $\hat{\eta}^{m,g}\hat{\xi}^{m,g}$ $(m = 1, 2, 3, 4)$ and $\hat{\zeta}^{g,full}$ from equation 6 estimated for each of the 19 bins separately for female (Panel A) and male (Panel B) students, with 50% as the baseline. The red area shows fitting in, the yellow area shows finding a role model, the green area shows doing well in studies, the blue area shows finding a career to pursue, and the purple area shows reasons other than these four.

50% in mathematics abilities in the sample, and Panels E and F restrict the sample to students among the top 50% in reading abilities in the sample.

Panel A shows that female students' preferences for gender composition are similar in STEM and non-STEM programs. Panel B reveals a different pattern for male students: they are less sensitive to gender composition in STEM than in non-STEM. Being a minority (75% female) reduces male students' choice probability by 8 percentage points in STEM versus 12 percentage points in non-STEM; being a majority (25% female) reduces it by 6 percentage points in STEM versus 8 percentage points in non-STEM.

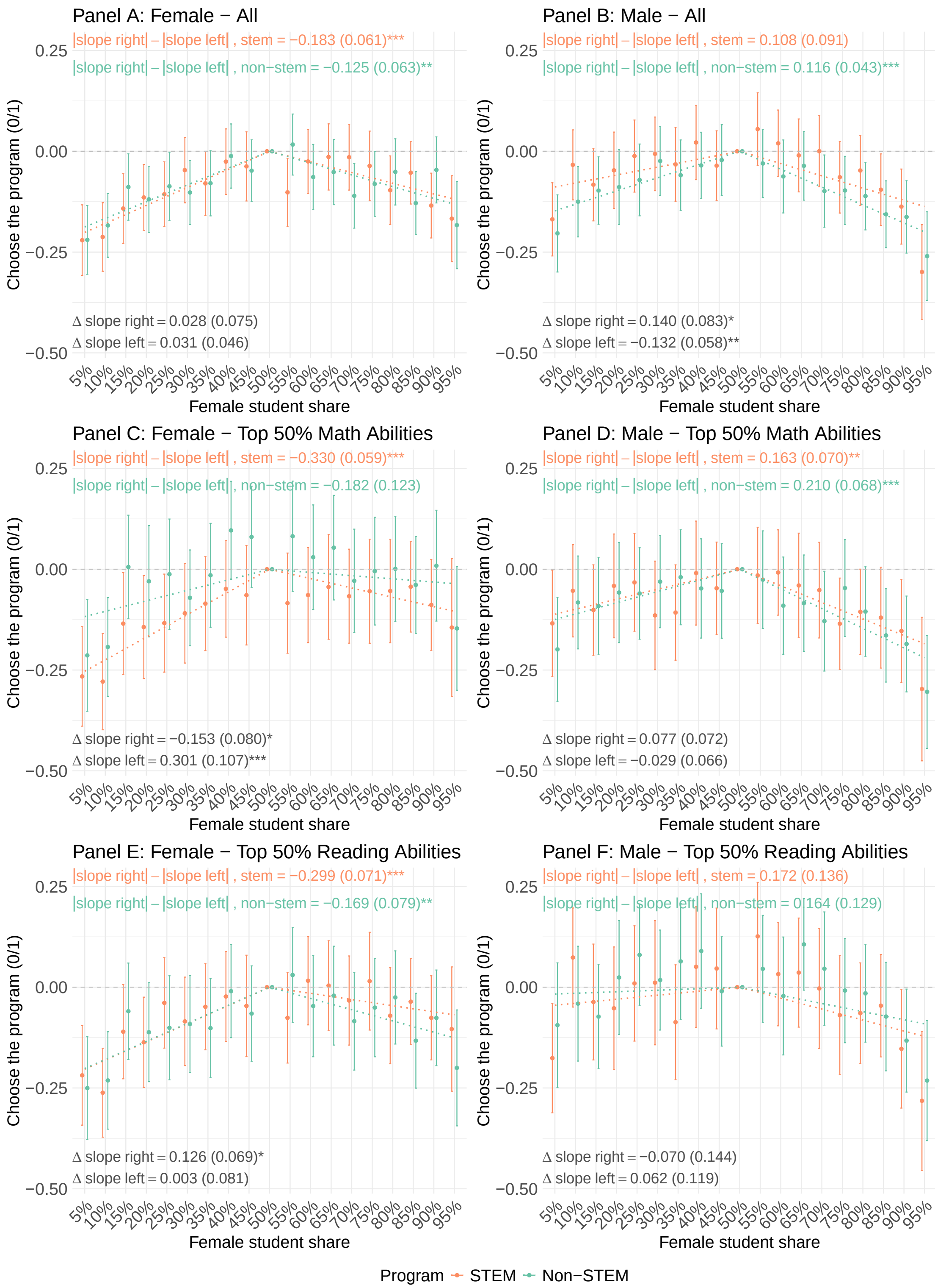
Panels C and D restrict the sample to students in the top half of mathematics ability. Here, female students exhibit stronger minority avoidance in STEM than non-STEM. Relative to a gender-balanced program, being a minority (25% female) reduces high-math-ability female students' choice probability by 14 percentage points in STEM versus 7 percentage points in non-STEM – a twofold difference. High-math-ability male students show no significant STEM/non-STEM heterogeneity (Panel D).[27]

Panels E and F examine students in the top half of reading ability; neither gender shows significant heterogeneity by field. Preferences for other attributes also do not differ between STEM and non-STEM (see Appendix Figure B4).

In summary, both genders prefer balance to majority and majority to minority. But sensitivity to gender composition varies by gender, ability, and field: high-mathematics-ability female students

27. Appendix Figure B2 presents differences in slopes between students in the top and bottom 50% of mathematics and reading abilities. These students also differ in their preferences for other attributes, as shown in Appendix Figure B3.

Figure 5: Heterogeneity of Preferences for the Gender Ratio – STEM vs. Non-STEM



Notes: This figure plots coefficient estimates and the 95% confidence intervals for female and male students, where we interact the attributes with the STEM dummy, with 50% as the baseline for both programs. Panels A and B plot all female and male students' preferences, Panels C and D plot female and male students in the top 50% of mathematics ability in the sample, and Panels E and F plot female and male students in the top 50% of reading ability in the sample. Standard errors are clustered at the student level. The dotted lines are weighted least squares linear fits of each point, with inverse variance weighting, for each side (below 50% and above 50%) for two groups. We imposed the constraint that the lines pass through the 50% point. The differences in the slopes between STEM and non-STEM are calculated using the weighted least squares standard errors. Significance levels: * 10%, ** 5%, and *** 1%.

are more deterred by minority status in STEM than in non-STEM, while male students are less sensitive to composition in STEM overall. One interpretation is that students perceive STEM as a male domain where male students feel they belong regardless of the gender ratio, whereas female students – especially those accustomed to male-dominated academic environments – recognize that fitting in requires a critical mass of same-gender peers.

6 Quantification of Inefficiency in Talent Allocation

Section 5.4 documents that preferences vary by gender, ability, and field. High-mathematics-ability female students exhibit stronger minority avoidance in STEM than in non-STEM, while male students show the opposite pattern – weaker sensitivity to gender composition in STEM overall. These preferences, combined with the low female share in STEM, imply potential inefficiency in talent allocation. This section quantifies that inefficiency.

We compare two scenarios in terms of how talent allocation changes between them:

- The optimal scenario: students have no preferences for the gender ratio. We use this as a benchmark for the best attainable allocation.[28]
- The baseline scenario: students have preferences for the gender ratio as estimated from the data, and the female share is 25% in STEM and 50% in non-STEM. We use this as a realistic baseline.

This comparison requires estimates of each student's preferences for program attributes, especially for the female share and STEM. We estimate these using a mixed logit model.

6.1 Econometric Framework

Estimation of Individual-Level Preferences Rewrite equation 2 as follows:

$$P^g_{ijd} = \frac{\exp(X'_{jd}\beta^g)}{\sum_{k\in\{r,l\}}\exp(X'_{jk}\beta^g)} \tag{7}$$

where P^g_{ijd} is student i of gender g's choice probability of program $d \in \{r, l\}$ with attributes X in pair j. This expression is equivalent to equation 2 in Section 5 (see, for example, Train 2009, Section 3.1). Unlike equation 2, however, we allow interactions between STEM and the female share to capture the heterogeneity documented in Section 5.4. We use a quadratic functional form for the female share to keep the number of parameters adequate for the sample size and to define choice probabilities over the continuous female share.

Now, let $\beta^g_i = (\alpha^g_i, \gamma^g)'$ and assume α^g_i is a random variable with density $f(\alpha^g|\theta^g)$, where θ^g are parameters of this distribution for gender g. We keep γ^g as fixed. Partition $X_{jd} = (X^1_{jd}, X^2_{jd})'$, where $X^1_{jd} = (STEM_{jd}, FShare_{jd}, FShare^2_{jd}, STEM_{jd} \times FShare_{jd}, STEM_{jd} \times FShare^2_{jd})'$. Then

28. Setting the STEM female share to 50% instead of removing gender preferences yields essentially the same results.

the choice probability can be written in a mixed logit form:[29]

$$P^g_{ijd}|\theta^g = \int \frac{\exp((X^{1\prime}_{jd}, X^{2\prime}_{jd})(\alpha^g_i, \gamma^g)')}{\sum_{k\in\{r,l\}} \exp((X^{1\prime}_{jk}, X^{2\prime}_{jk})(\alpha^g_i, \gamma^g)')} f(\alpha^g|\theta^g) d\alpha^g \tag{8}$$

where we allow arbitrary correlations among elements in α^g_i except the parameter for STEM. We assume that the mixing distribution f follows a normal distribution, as is standard (Train 2009).[30] Continuous variables are normalized for model stability.

We estimate θ^g by maximum simulated likelihood, approximating the integral using 4,000 Halton draws for stability and faster convergence (Section 9). We then obtain individual-specific parameters α^g_i as posterior means of their conditional distribution, following Train (2009):

$$h(\alpha^g|y_i, X^1_i, X^2_i, g, \theta^g) \tag{9}$$

where y_i is a vector of i's choices across the 15 pairs, and X^1_i and X^2_i are matrices of attributes in X^1_{jd} and X^2_{jd} that i observed across the 15 pairs.[31]

6.2 Utility Cost

We first examine the utility cost of a low female share in STEM among students who prefer STEM. Panel A of Appendix Figure B5 shows that, compared to the optimal scenario, female students incur a utility cost of 0.39 and male students 0.15 under the baseline scenario. As shown in Panel B, this is equivalent to 31.6% of the STEM preference for female students and 9.8% for male students.[32] This is a substantial cost: 15.2% of female students and 4.0% of male students forgo STEM due to the low female share.

6.3 Inefficiency in Talent Allocation

We now examine how the low female share in STEM affects talent allocation. Since both female and male students incur utility costs from a low female share in STEM, removing gender-ratio preferences in the optimal scenario allows more students to enter STEM. However, when enrollment differs across scenarios, comparing ability sorting becomes difficult. To make a meaningful comparison, we impose capacity constraints in the optimal scenario, setting each field's capacity equal to its

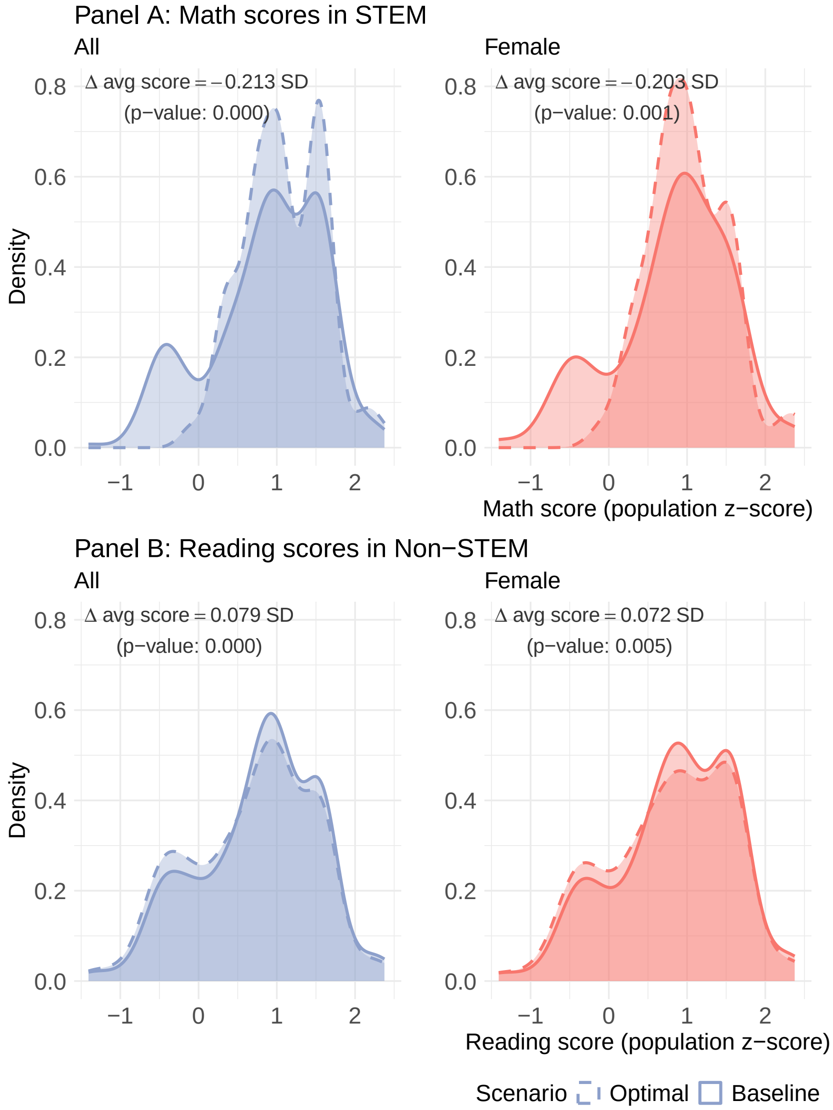
29. Mixed logit relaxes the following standard logit assumptions: (i) no random preference variation among individuals, (ii) independence of irrelevant alternatives, and (iii) no correlation in unobserved factors over time (Train 2009, Section 6). Our logit results in Table 2 remain valid as there are only two alternatives in each pair and no notable time-varying unobserved factors in the experiment. We use mixed logit to relax the first assumption and estimate individual-level preference parameters.

30. Appendix Section A discusses the mixed logit model performance and variation in individual-level preference parameters.

31. Individual-specific parameters from mixed logit are used in other papers. For example, León and Miguel (2017) estimate the value of a statistical life among Africans and non-Africans using individual-level parameters. Kremer et al. (2011) estimate the value of spring water protection in Kenyan households using individual-level parameters.

32. Panel B excludes the bottom 10% of students whose STEM preference is weak, as they inflate the willingness to pay.

Figure 6: Changes in the Ability Distributions



Notes: This figure plots students' ability distributions under the baseline scenario (solid) and the optimal scenario (dashed). Panel A shows the distributions for STEM and Panel B for non-STEM. Gray plots show distributions for all students, red for female students, and green for male students. The change in average scores from the optimal to the baseline scenarios, along with bootstrapped p-values, is presented in the top left of each plot.

enrollment under the baseline scenario. We allocate students using the student-proposing deferred acceptance algorithm (Gale and Shapley 1962).[33]

The algorithm works as follows:

1. Students apply to their most preferred field (STEM or non-STEM) according to their preferences.

33. The student-proposing deferred acceptance algorithm is strategy-proof and stable. Although alternative algorithms such as the Boston mechanism and Top Trading Cycles often yield higher welfare, the gains typically arise from institutional or cognitive constraints (see, e.g., Calsamiglia, Fu, and Güell 2020; Calsamiglia, Haeringer, and Klijn 2010).

2. Each field ranks students by subject-specific ability – mathematics scores for STEM, reading scores for non-STEM – and tentatively admits the top-ranked students up to capacity, rejecting the rest.
3. Rejected students apply to their next preferred field.

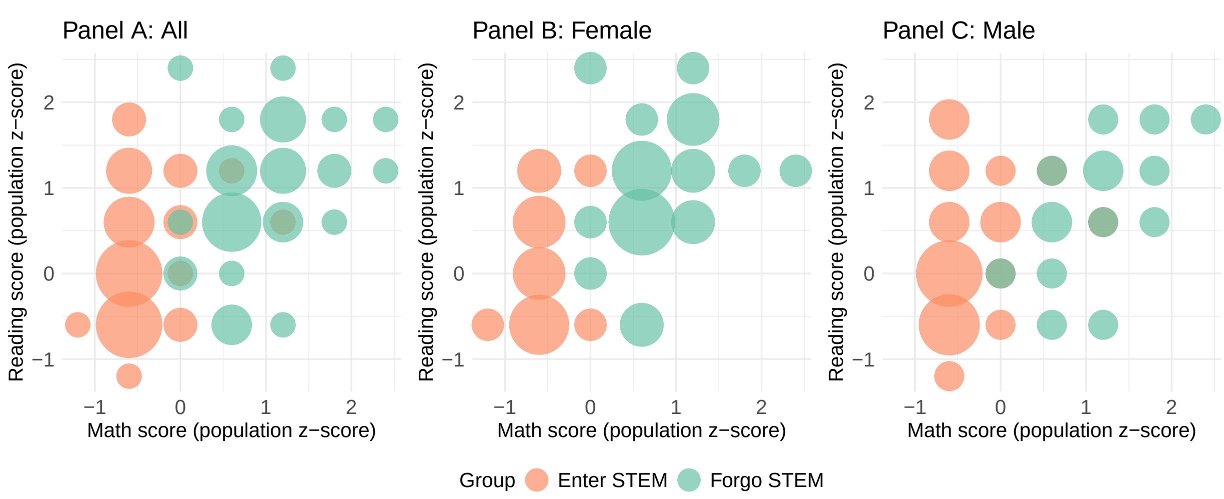
Steps 2 and 3 repeat until no further rejections occur. Capacity for each field is set equal to the number of students allocated to that field under the baseline scenario without capacity constraints (i.e., when all students are admitted to their most preferred field).

Figure 6 plots students' ability distributions under the optimal scenario (dashed) and the baseline scenario (solid). Panel A shows distributions for STEM and Panel B for non-STEM. Gray plots show all students, red female students, and green male students. Changes in average scores from the optimal to the baseline scenario, along with bootstrapped p-values, are presented in the top left of each plot.

Panel A shows that the average STEM student's mathematics score is 0.2 standard deviations lower under the baseline than the optimal scenario. This reflects inefficient talent allocation: some female students scoring 1 standard deviation above the population mean, and some male students scoring 1–1.5 standard deviations above the mean, shift to non-STEM under the baseline. Their seats are filled by students scoring 0.5–1 standard deviations below the mean.

Panel B shows that the average non-STEM student's reading score is 0.08 standard deviations higher under the baseline – allocation in non-STEM is thus less efficient under the optimal scenario. However, the efficiency loss in non-STEM is only one-third the size of the efficiency gain in STEM, so overall allocation improves.

Figure 7: Students Who Forgo and Enter STEM Due to Low Female Share



Notes: This figure plots students who forgo (green dots) and enter (orange dots) STEM due to the low female share. Dot size indicates the number of students in each cell.

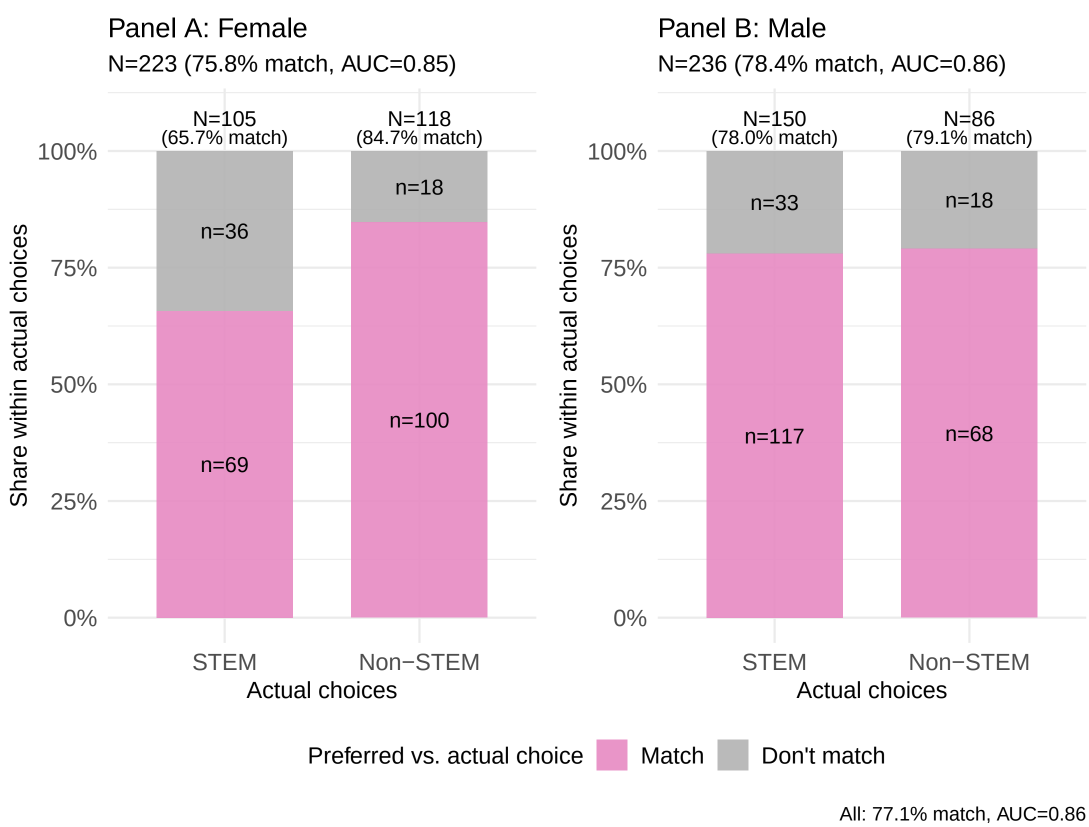
To better understand which students forgo and enter STEM due to the low female share, Figure 7 plots both groups. Dot size indicates the number of students in each cell.

Panel A shows that students who forgo STEM lie roughly on the 45-degree line – they excel in both mathematics and reading – while those who enter do not excel in mathematics. Panels B and C show similar patterns by gender, though more female students forgo STEM than male students, consistent with the stronger minority avoidance in STEM among high-mathematics-ability female students documented in Section 5.4. Thus, the inefficiency arises because students who excel in both subjects – and hence have more freedom to choose – shift to non-STEM, while their seats are filled by students who do not excel in mathematics.

In summary, the low female share in STEM leads to inefficient talent allocation: students who excel in both mathematics and reading shift from STEM to non-STEM, with female students more likely to forgo, while students who do not excel in mathematics fill their places. The resulting efficiency loss in STEM is only partially offset by a modest efficiency gain in non-STEM.

7 Preferences and Actual Choices

Figure 8: Preferences and Actual Choices



Notes: This figure shows the fraction of students who chose their preferred track. We classify a student's preferred track as STEM if the mixed logit coefficient estimate for the STEM dummy is positive, and non-STEM otherwise. We denote the sciences track as STEM and the humanities track as non-STEM.

Finally, we examine whether students choose their preferred track. Figure 8 shows the results.

We classify a student's preferred track as STEM if the mixed logit coefficient for the STEM dummy is positive, and non-STEM otherwise; we denote the sciences track as STEM and the humanities track as non-STEM.

Overall, 77.1% of students chose their preferred track (75.8% for female students, 78.4% for male students). We assess classification quality using the ROC-AUC, which measures the model's ability to discriminate between STEM and non-STEM choosers (Hanley and McNeil 1982). The AUC indicates excellent discriminatory power (Hosmer and Lemeshow 2000, Chapter 5): 0.86 for all students, 0.86 for male students, and 0.85 for female students. This is notable given that AUC analysis typically evaluates contemporaneous signals and outcomes, whereas our data span several months between the experiment and actual track choices.

Second, female students who prefer STEM are 12.3 percentage points less likely to follow through than male students who prefer STEM. This gap is broadly consistent in magnitude with the estimated fraction of students who forgo STEM due to the low female share (Panel B of Appendix Figure B5). Importantly, this gap does not reflect differential prediction accuracy: we classify preferences using only the coefficient on the STEM dummy, which captures intrinsic field preferences but not preferences over gender composition. Since female share preferences are excluded from the classification, female students classified as "preferring STEM" may still avoid it due to the low female share – consistent with our main finding that female students anticipate difficulty fitting into male-dominated environments.

8 Discussion and Conclusion

Female students are significantly less likely to pursue STEM fields at colleges in OECD countries, despite the negligible gender gap in mathematics and sciences at age 15. One potential reason that has received little attention is that STEM programs are male-dominated, which may discourage female students from pursuing STEM in anticipation of the disadvantages of being a gender minority. In this paper, we examine whether the gender ratio affects students' college choices and whether the low female share in STEM leads to an inefficient allocation of talent, using an incentivized discrete choice experiment with high school students.

We find that the gender ratio affects the college choices of both female and male students. Both genders prefer programs with a balanced gender ratio over those dominated by either gender, and both prefer being a majority to being a minority. A decomposition reveals that students avoid being a minority primarily because they anticipate difficulty fitting in. Importantly, these preferences vary by context: female students with high mathematics ability are particularly averse to being a minority in STEM programs, while male students are generally less sensitive to the gender ratio in STEM than in non-STEM programs. A quantification exercise shows that the low female share in STEM, coupled with these preferences, generates inefficiency in talent allocation – high-mathematics-ability students are deterred from entering STEM, and their seats are filled by low-mathematics-ability students, with effects more pronounced among female students. Finally, using actual choice data collected several months later, we confirm that most students pursue fields consistent with their experimentally elicited preferences, but female students who prefer STEM are less likely to follow through – consistent with their stronger aversion to being a minority.

Our findings carry several implications. First, they suggest that the low female share in STEM is self-perpetuating: the current gender imbalance deters the very students who might otherwise enter, reinforcing the imbalance. This creates a coordination problem where individual students' rational responses to the existing composition collectively sustain an inefficient equilibrium. Second, our results provide an efficiency-based rationale for policies aimed at increasing female representation in STEM. Interventions that create more welcoming environments for female students can be justified not only on equity grounds but also because they may improve talent allocation.

Third, the finding that anticipated difficulty fitting in drives minority avoidance points toward specific mechanisms that interventions might target – for instance, providing information about peer support or highlighting successful integration of minority students. Such initiatives would benefit society beyond improving talent allocation: they can help address gender differences in occupational sorting and narrow the gender wage gap, shifting social norms toward greater gender equality in the long run. More broadly, more gender-equal societies benefit all members. In professional settings, they foster better recognition of ideas (Koffi 2025) and improve team performance (Hoogendoorn, Oosterbeek, and van Praag 2013). In social life, they reduce hostility toward same-sex marriage (Baranov, De Haas, and Grosjean 2023) and support higher rates of cross-gender friendships (Bailey et al. 2025).

Several limitations warrant discussion. Our sample consists of high-achieving students at selective high schools in the Greater Tokyo area, who face fewer constraints on their college choices than students elsewhere in Japan or in other countries. Whether our findings generalize to students with weaker mathematics ability or tighter geographic constraints remains an open question. Additionally, while our incentivized choice experiment provides clean identification of preferences, it cannot capture all factors that influence actual college decisions, including parental pressure, teacher recommendations, and peer influences.

Future research could extend this work in several directions. Examining whether preferences for gender balance vary across countries with different cultural norms around gender would help assess the generalizability of our findings. Investigating whether information interventions – such as highlighting successful female STEM students or providing realistic previews of the social environment—can attenuate minority avoidance would have direct policy relevance. Finally, tracking students longitudinally to examine whether those who enter gender-imbalanced programs experience the difficulties they anticipate would shed light on whether students' beliefs are accurately calibrated.

In conclusion, our findings suggest that the gender composition of STEM programs is not merely a symptom of other factors driving female students away, but is itself a barrier to entry. Breaking this cycle may require coordinated efforts to shift the equilibrium. Small increases in female representation could reduce the cost of entry for subsequent cohorts, potentially generating momentum toward greater balance. More broadly, our results underscore how compositional features of educational environments can shape who pursues which fields, with consequences for both individual careers and the efficient allocation of talent in society.

References

Araki, Shota, Daiji Kawaguchi, and Yuki Onozuka. 2016. "University Prestige, Performance Evaluation, and Promotion: Estimating the Employer Learning Model Using Personnel Datasets." *Labour Economics,* SOLE/EALE Conference Issue 2015, 41:135–148.

Artmann, Elisabeth, Nadine Ketel, Hessel Oosterbeek, and Bas van der Klaauw. 2018. *Field of Study and Family Outcomes.* Working Paper 11658. Institute of Labor Economics (IZA).

———. 2021. "Field of Study and Partner Choice." *Economics of Education Review* 84:102149.

Avery, Mallory, Jane Caldwell, Christian D. Schunn, and Katherine Wolfe. 2024. "Improving Introductory Economics Course Content and Delivery Improves Outcomes for Women." *The Journal of Economic Education* 55 (3): 216–231.

Bailey, Michael, Drew Johnston, Theresa Kuchler, Ayush Kumar, and Johannes Stroebel. 2025. "Cross-Gender Social Ties around the World." *AEA Papers and Proceedings* 115:132–138.

Baltrunaite, Audinga, Piera Bello, Alessandra Casarico, and Paola Profeta. 2014. "Gender Quotas and the Quality of Politicians." *Journal of Public Economics* 118:62–74.

Baranov, Victoria, Ralph De Haas, and Pauline Grosjean. 2023. "Men. Male-biased Sex Ratios and Masculinity Norms: Evidence from Australia's Colonial Past." *Journal of Economic Growth* 28 (3): 339–396.

Bechichi, Nagui, and Gustave Kenedi. 2024. *Older Schoolmate Spillovers on Higher Education Choices.* Working Paper.

Bello, Piera, Alessandra Casarico, and Debora Nozza. 2025. "Research Similarity and Women in Academia." *The Economic Journal,* ueaf113.

Besley, Timothy, Olle Folke, Torsten Persson, and Johanna Rickne. 2017. "Gender Quotas and the Crisis of the Mediocre Man: Theory and Evidence from Sweden." *American Economic Review* 107 (8): 2204–2242.

Booth, Alison L., Lina Cardona-Sosa, and Patrick Nolen. 2018. "Do Single-Sex Classes Affect Academic Achievement? An Experiment in a Coeducational University." *Journal of Public Economics* 168:109–126.

Bordon, Paola, and Chao Fu. 2015. "College-Major Choice to College-Then-Major Choice." *The Review of Economic Studies* 82 (4): 1247–1288.

Bostwick, Valerie K., and Bruce A. Weinberg. 2022. "Nevertheless She Persisted? Gender Peer Effects in Doctoral STEM Programs." *Journal of Labor Economics* 40 (2): 397–436.

Breda, Thomas, Julien Grenet, Marion Monnet, and Clémentine Van Effenterre. 2023. "How Effective Are Female Role Models in Steering Girls Towards STEM? Evidence from French High Schools." *The Economic Journal* 133 (653): 1773–1809.

Breda, Thomas, and Clotilde Napp. 2019. "Girls' Comparative Advantage in Reading Can Largely Explain the Gender Gap in Math-Related Fields." *Proceedings of the National Academy of Sciences* 116 (31): 15435–15440.

Burbano, Vanessa, Nicolas Padilla, and Stephan Meier. 2024. "Gender Differences in Preferences for Meaning at Work." *American Economic Journal: Economic Policy* 16 (3): 61–94.

Buser, Thomas, Muriel Niederle, and Hessel Oosterbeek. 2014. "Gender, Competitiveness, and Career Choices." *The Quarterly Journal of Economics* 129 (3): 1409–1447.

———. 2024. "Can Competitiveness Predict Education and Labor Market Outcomes? Evidence from Incentivized Choice and Survey Measures." *The Review of Economics and Statistics,* 1–45.

Calsamiglia, Caterina, Chao Fu, and Maia Güell. 2020. "Structural Estimation of a Model of School Choices: The Boston Mechanism versus Its Alternatives." *Journal of Political Economy* 128 (2): 642–680.

Calsamiglia, Caterina, Guillaume Haeringer, and Flip Klijn. 2010. "Constrained School Choice: An Experimental Study." *American Economic Review* 100 (4): 1860–1874.

Canaan, Serena, and Pierre Mouganie. 2023. "The Impact of Advisor Gender on Female Students' STEM Enrollment and Persistence." *Journal of Human Resources* 58 (2): 593–632.

Carlana, Michela. 2019. "Implicit Stereotypes: Evidence from Teachers' Gender Bias." *The Quarterly Journal of Economics* 134 (3): 1163–1224.

Carlana, Michela, and Lucia Corno. 2024. "Thinking about Parents: Gender and Field of Study." *AEA Papers and Proceedings* 114:254–258.

———. 2025. *Peer Influence in Educational Choices: Social Image Concerns and Same-Gender Interactions.* Working Paper.

Carrell, Scott E., Marianne E. Page, and James E. West. 2010. "Sex and Science: How Professor Gender Perpetuates the Gender Gap." *Quarterly Journal of Economics* 125 (3): 1101–1144.

Chan, Alex. 2024. *Discrimination Against Doctors: A Field Experiment.* Working Paper.

Croson, Rachel, and Uri Gneezy. 2009. "Gender Differences in Preferences." *Journal of Economic Literature* 47 (2): 448–474.

Di Tommaso, Maria Laura, Dalit Contini, Dalila De Rosa, Francesca Ferrara, Daniela Piazzalunga, and Ornella Robutti. 2024. "Tackling the Gender Gap in Mathematics with Active Learning Methodologies." *Economics of Education Review* 100:102538.

Dohmen, Thomas, Armin Falk, David Huffman, Uwe Sunde, Jürgen Schupp, and Gert G. Wagner. 2011. "Individual Risk Attitudes: Measurement, Determinants, and Behavioral Consequences." *Journal of the European Economic Association* 9 (3): 522–550.

Einiö, Elias, Josh Feng, and Xavier Jaravel. 2025. *Social Push and the Direction of Innovation.* Working Paper 3383703.

Ersoy, Fulya, and Jamin D. Speer. 2025. "Opening the Black Box of College Major Choice: Evidence from an Information Intervention." *Journal of Economic Behavior & Organization* 231:106800.

Fischer, Stefanie. 2017. "The Downside of Good Peers: How Classroom Composition Differentially Affects Men's and Women's STEM Persistence." *Labour Economics* 46:211–226.

Folke, Olle, and Johanna Rickne. 2022. "Sexual Harassment and Gender Inequality in the Labor Market." *The Quarterly Journal of Economics* 137 (4): 2163–2212.

Funk, Patricia, Nagore Iriberri, and Giulia Savio. 2024. "Does Scarcity of Female Instructors Create Demand for Diversity among Students? Evidence from an M-Turk Experiment." *Labour Economics* 90:102606.

Gale, David, and Lloyd S. Shapley. 1962. "College Admissions and the Stability of Marriage." *The American Mathematical Monthly* 69 (1): 9–15.

Gallen, Yana, and Melanie Wasserman. 2023. "Does Information Affect Homophily?" *Journal of Public Economics* 222:104876.

Gelbach, Jonah B. 2016. "When Do Covariates Matter? And Which Ones, and How Much?" *Journal of Labor Economics* 34 (2): 509–543.

Genda, Yuji, Ayako Kondo, and Souichi Ohta. 2010. "Long-Term Effects of a Recession at Labor Market Entry in Japan and the United States." *Journal of Human Resources* 45 (1): 157–196.

Giustinelli, Pamela. 2016. "Group Decision Making with Uncertain Outcomes: Unpacking Child–Parent Choice of the High School Track." *International Economic Review* 57 (2): 573–602.

Gong, Jie, Yi Lu, and Hong Song. 2021. "Gender Peer Effects on Students' Academic and Noncognitive Outcomes: Evidence and Mechanisms." *Journal of Human Resources* 56 (3): 686–710.

Gong, Yifan, Lance Lochner, Ralph Stinebrickner, and Todd R. Stinebrickner. 2021. *The Consumption Value of College.* Working Paper.

Goulas, Sofoklis, Silvia Griselda, and Rigissa Megalokonomou. 2024. "Comparative Advantage and Gender Gap in STEM." *Journal of Human Resources* 59 (6): 1937–1980.

Hainmueller, Jens, Dominik Hangartner, and Teppei Yamamoto. 2015. "Validating Vignette and Conjoint Survey Experiments against Real-World Behavior." *Proceedings of the National Academy of Sciences* 112 (8): 2395–2400.

Hampole, Menaka, Francesca Truffa, and Ashley Wong. 2024. *Peer Effects and the Gender Gap in Corporate Leadership: Evidence from MBA Students.* Working Paper.

Hanley, James A., and Barbara J. McNeil. 1982. "The Meaning and Use of the Area under a Receiver Operating Characteristic (ROC) Curve." *Radiology* 143 (1): 29–36.

Högn, Celina, Lea Mayer, Johannes Rincke, and Erwin Winkler. 2025. *Preferences for Gender Diversity in High-Profile Jobs.* Working Paper.

Hoogendoorn, Sander, Hessel Oosterbeek, and Mirjam van Praag. 2013. "The Impact of Gender Diversity on the Performance of Business Teams: Evidence from a Field Experiment." *Management Science* 59 (7): 1514–1528.

Hosmer, David W., and Stanley Lemeshow. 2000. *Applied Logistic Regression.* New York, NY: Wiley.

Inoue, Chihiro, Asumi Saito, and Yuki Takahashi. 2023. "STEM Gender Gap." AEA RCT Registry. December 20. https://doi.org/10.1257/rct.12577-1.0.

Jacob, Brian, Brian McCall, and Kevin Stange. 2018. "College as Country Club: Do Colleges Cater to Students' Preferences for Consumption?" *Journal of Labor Economics* 36 (2): 309–348.

Karpowitz, Christopher F., Stephen D. O'Connell, Jessica Preece, and Olga Stoddard. 2024. "Strength in Numbers? Gender Composition, Leadership, and Women's Influence in Teams." *Journal of Political Economy* 132 (9): 3077–3114.

Kessler, Judd B., Corinne Low, and Colin D. Sullivan. 2019. "Incentivized Resume Rating: Eliciting Employer Preferences without Deception." *American Economic Review* 109 (11): 3713–3744.

Kirkebøen, Lars, Edwin Leuven, Magne Mogstad, and Jack Mountjoy. 2025. *College as a Marriage Market.* Working Paper 28688. National Bureau of Economic Research.

Koffi, Marlène. 2025. "Innovative Ideas and Gender (In)Equality." *American Economic Review* 115 (7): 2207–2236.

Koning, Rembrand, Sampsa Samila, and John-Paul Ferguson. 2021. "Who Do We Invent for? Patents by Women Focus More on Women's Health, but Few Women Get to Invent." *Science* 372 (6548): 1345–1348.

Kremer, Michael, Jessica Leino, Edward Miguel, and Alix Peterson Zwane. 2011. "Spring Cleaning: Rural Water Impacts, Valuation, and Property Rights Institutions." *The Quarterly Journal of Economics* 126 (1): 145–205.

León, Gianmarco, and Edward Miguel. 2017. "Risky Transportation Choices and the Value of a Statistical Life." *American Economic Journal: Applied Economics* 9 (1): 202–228.

List, John A., and Jason F. Shogren. 1998. "Calibration of the Difference between Actual and Hypothetical Valuations in a Field Experiment." *Journal of Economic Behavior & Organization* 37 (2): 193–205.

List, John A., Paramita Sinha, and Michael H. Taylor. 2006. "Using Choice Experiments to Value Non-Market Goods and Services: Evidence from Field Experiments." *The B.E. Journal of Economic Analysis & Policy* 6 (2).

Long, Dede, and Yuki Takahashi. 2025. *Can Curricular Reform Close the STEM Gender Gap? Evidence from an Introductory Computer Science Course.* Working Paper.

Low, Corinne. 2024. "Pricing the Biological Clock: The Marriage Market Costs of Aging to Women." *Journal of Labor Economics* 42 (2): 395–426.

Macchi, Elisa. 2023. “Worth Your Weight: Experimental Evidence on the Benefits of Obesity in Low-Income Countries.” *American Economic Review* 113 (9): 2287–2322.

Ministry of Education, Culture, Sports, Science and Technology. 2021. *On the Current State of High School Education [Original in Japanese].* Report.

Miserocchi, Francesca. 2024. *Discrimination through Biased Memory.* Working Paper.

Moriguchi, Chiaki. 2014. “Japanese-Style Human Resource Management and Its Historical Origins.” *Japan Labor Review* 11 (3): 58–77.

Mouganie, Pierre, and Yaojing Wang. 2020. “High-Performing Peers and Female STEM Choices in School.” *Journal of Labor Economics* 38 (3): 805–841.

Müller, Maximilian W. 2024. *Parental Pressure and Educational Choices.* Working Paper.

Nakajima, Koji. 2018. “Analysis on Deviation Value and Employment in Major Companies: Reconsideration of System for Recruiting New Graduates [Original in Japanese].” *Kansai University Journal of Higher Education* 9:57–68.

National Federation of University Cooperative Associations. 2023. *Preliminary Report of the 58th Student Life Survey [Original in Japanese].* Report. National Federation of University Co-operative Associations.

OECD. 2019. *PISA 2018 Results: Where All Students Can Succeed.* Vol. 2. Paris, France: OECD Publishing.

Owen, Ann L., and Paul Hagstrom. 2021. “Broadening Perceptions of Economics in a New Introductory Economics Sequence.” *The Journal of Economic Education* 52 (3): 175–191.

Pestel, Nico. 2021. “Searching on Campus? The Marriage Market Effects of Changing Student Sex Ratios.” *Review of Economics of the Household* 19 (4): 1175–1207.

Porter, Catherine, and Danila Serra. 2020. “Gender Differences in the Choice of Major: The Importance of Female Role Models.” *American Economic Journal: Applied Economics* 12 (3): 226–254.

Riise, Julie, Barton Willage, and Alexander Willén. 2022. “Can Female Doctors Cure the Gender STEMM Gap? Evidence from Exogenously Assigned General Practitioners.” *The Review of Economics and Statistics* 104 (4): 621–635.

Riley, Emma. 2024. “Role Models in Movies: The Impact of Queen of Katwe on Students’ Educational Attainment.” *The Review of Economics and Statistics* 106 (2): 334–351.

Schuh, Rachel. 2024. *Miss-Allocation: The Value of Workplace Gender Composition and Occupational Segregation.* Working Paper.

Shan, Xiaoyue. 2024. *Gender Diversity Improves Academic Performance.* Working Paper.

Tadjfar, Nagisa, and Kartik Vira. 2025. *Friends in Higher Places: Social Fit and University Choice.* Working Paper.

The Economist. 2024. “Why Young Men and Women Are Drifting Apart.” *The Economist.*

Train, Kenneth E. 2009. *Discrete Choice Methods with Simulation.* 2nd ed. Cambridge, UK: Cambridge University Press.

Truffa, Francesca, and Ashley Wong. 2025. "Undergraduate Gender Diversity and the Direction of Scientific Research." *American Economic Review* 115 (7): 2414–2448.

Valdebenito, Rocío. 2023. *Peer Influence and College Major Choices in Male-Dominated Fields.* Working Paper.

Wiswall, Matthew, and Basit Zafar. 2018. "Preference for the Workplace, Investment in Human Capital, and Gender." *The Quarterly Journal of Economics* 133 (1): 457–507.

———. 2021. "Human Capital Investments and Expectations about Career and Family." *Journal of Political Economy* 129 (5): 1361–1424.

Zafar, Basit. 2013. "College Major Choice and the Gender Gap." *Journal of Human Resources* 48 (3): 545–595.

Zizzo, Daniel John. 2010. "Experimenter Demand Effects in Economic Experiments." *Experimental Economics* 13 (1): 75–98.

Online Appendix

A Mixed Logit Model Performance

Table A1 presents mixed logit parameter estimates for female students (column 1) and male students (column 2). The table shows substantial and well-identified heterogeneity in preferences for both groups. The standard deviation of the random STEM coefficient is 1.62 for females and 1.60 for males, both highly statistically significant ($p < 0.001$). This implies wide variation in baseline STEM preferences. Heterogeneity in preferences for female share also exhibits substantial variation: for females, the standard deviations are approximately 0.28 for the linear term and 0.22 for the quadratic term, while for males they are approximately 0.37 and 0.42, respectively.

The model fit is adequate for both samples: the log-likelihood is -2,747.5 for females and -2,632.2 for males, and McFadden's pseudo-R-squared is 0.147 for females and 0.146 for males. These values confirm that the data contain sufficient within-individual variation to identify the distribution of random coefficients and the associated individual-level parameters.

The estimated covariance matrices display structured and interpretable correlation patterns, as shown in Table A2. For example, the correlation between the linear and quadratic female share terms is -0.364 for females and 0.704 for males, consistent with heterogeneous nonlinear responses. Moreover, the correlation between STEM times female share and female share is 0.876 for females and -0.834 for males, indicating systematic rather than idiosyncratic interaction heterogeneity.

B Additional Figures and Tables

Table A1: Mixed Logit Parameter Estimates

Sample:	Female	Male
Outcome:	Choose the program (0/1)	
	(1)	(2)
STEM	-0.449***	0.218***
	(0.075)	(0.074)
Female student share	0.144***	-0.065
	(0.039)	(0.040)
Female student share squared	-0.282***	-0.349***
	(0.043)	(0.045)
Selectivity index (population SD)	0.153***	0.267***
	(0.026)	(0.028)
Cohort size/100	0.039	0.052*
	(0.026)	(0.028)
Intl exchange program	0.370***	0.177**
	(0.066)	(0.070)
Club participation rate	0.139***	0.090***
	(0.026)	(0.028)
STEM x Female student share	0.008	-0.011
	(0.056)	(0.056)
STEM x Female student share squared	-0.046	0.015
	(0.063)	(0.061)
Intercept for right	-0.008	-0.056
	(0.037)	(0.038)
SD(STEM)	1.618***	1.595***
	(0.078)	(0.080)
Chol(FShare, FShare)	0.281***	0.365***
	(0.076)	(0.079)
Chol(FShare, FShare squared)	-0.079	0.178**
	(0.067)	(0.072)
Chol(FShare squared, FShare squared)	0.203**	0.180**
	(0.081)	(0.085)
Chol(FShare, FShare x STEM)	0.009	-0.111
	(0.111)	(0.110)
Chol(FShare, FShare squared x STEM)	0.013	-0.264***
	(0.080)	(0.084)
Chol(FShare squared, FShare x STEM)	0.120	-0.000
	(0.097)	(0.095)
Chol(FShare squared, FShare squared x STEM)	0.091	0.152
	(0.103)	(0.135)
Chol(FShare x STEM, FShare x STEM)	0.029	-0.171
	(0.488)	(0.166)
Chol(FShare x STEM, FShare squared x STEM)	-0.068	-0.078
	(0.714)	(0.154)
Chol(FShare squared x STEM, FShare squared x STEM)	0.015	-0.025
	(3.719)	(2.056)
Log Likelihood	-2747.457	-2632.201
McFadden's Pseudo-R-squared	0.147	0.146
No. observations	9298	8902
No. students	310	297

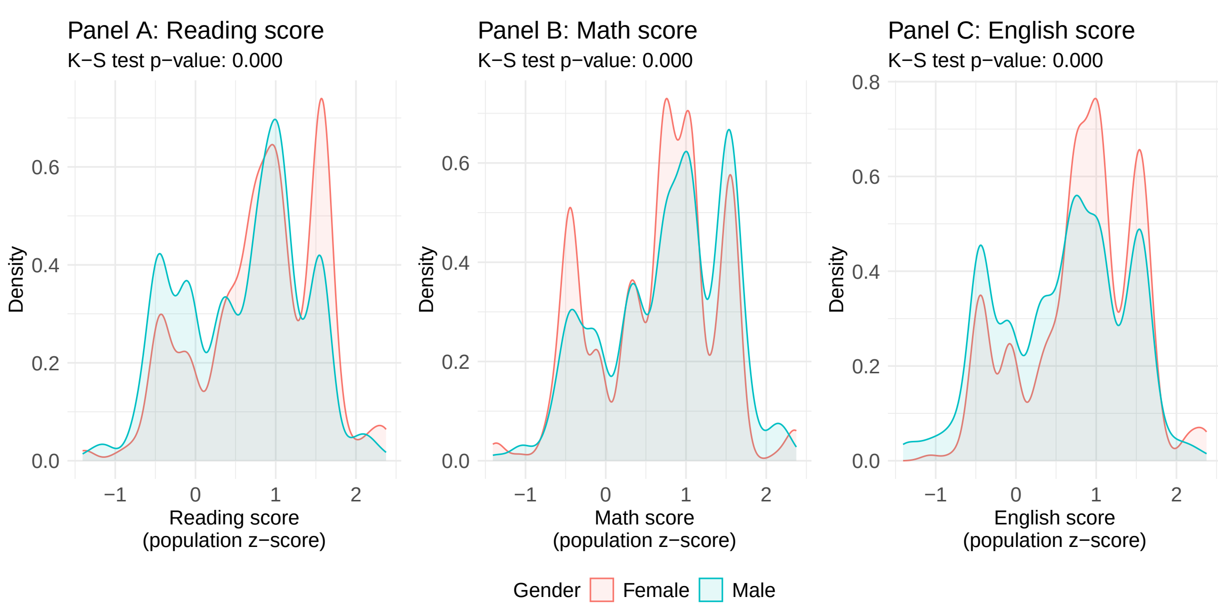
Notes: This table presents mixed logit parameter estimates for female students (column 1) and male students (column 2). Significance levels: * 10%, ** 5%, and *** 1%.

Table A2: Correlation Matrix of Mixed Logit Individual-Level Parameters

Panel A: Females				
	Female share	Female share squared	STEM x Female share	STEM x Female share squared
Female share	1.000	-0.364	0.074	0.117
Female share squared	-0.364	1.000	0.876	0.692
STEM x Female share	0.074	0.876	1.000	0.636
STEM x Female share squared	0.117	0.692	0.636	1.000
Panel B: Males				
	Female share	Female share squared	STEM x Female share	STEM x Female share squared
Female share	1.000	0.704	-0.543	-0.836
Female share squared	0.704	1.000	-0.384	-0.247
STEM x Female share	-0.543	-0.384	1.000	0.663
STEM x Female share squared	-0.836	-0.247	0.663	1.000

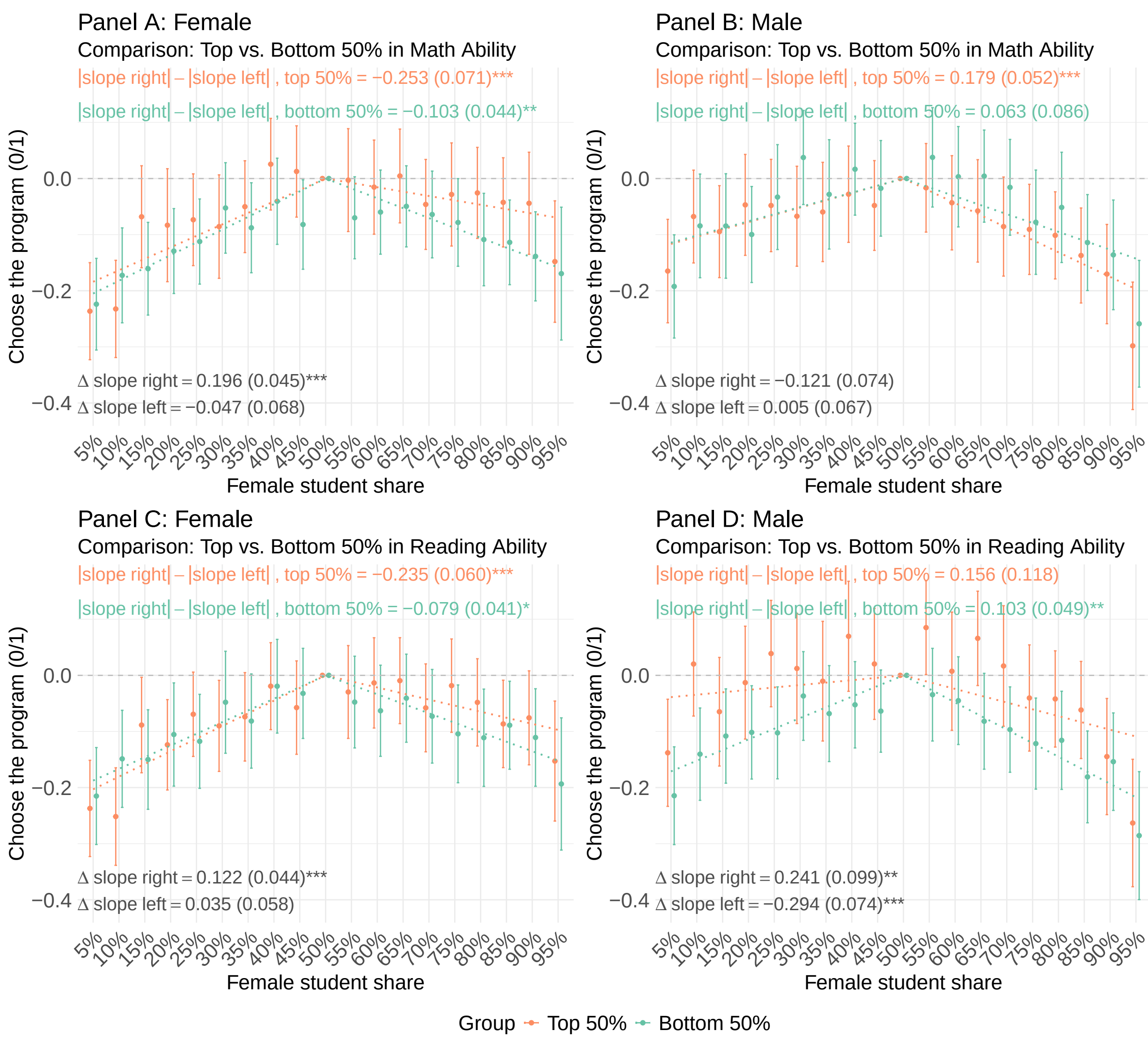
Notes: This table reports the correlation matrix of mixed logit individual-level parameters for female (Panel A) and male (Panel B) students.

Figure B1: Distribution of Abilities



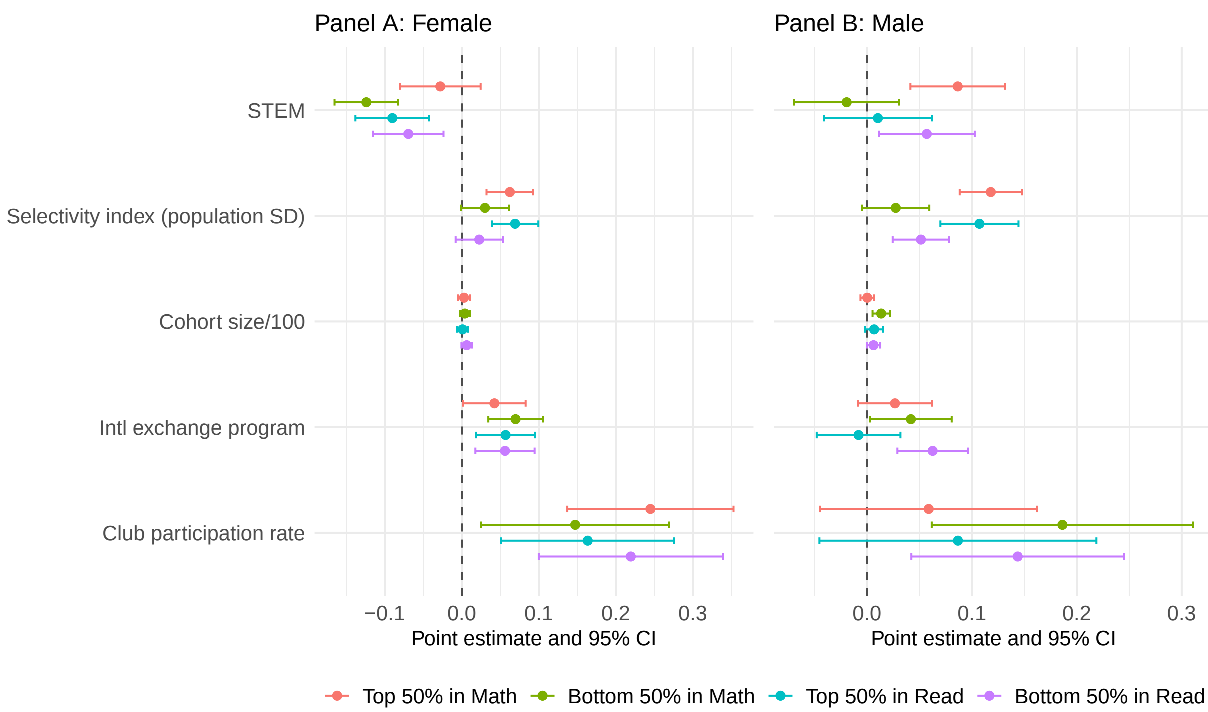
Notes: This figure presents the distribution of abilities of female and male students in reading (Panel A), mathematics (Panel B), and English (Panel C). K-S test p-value shows the Kolmogorov-Smirnov test p-value for differences in the distribution between female and male students.

Figure B2: Heterogeneity of Preferences for the Gender Ratio by Abilities



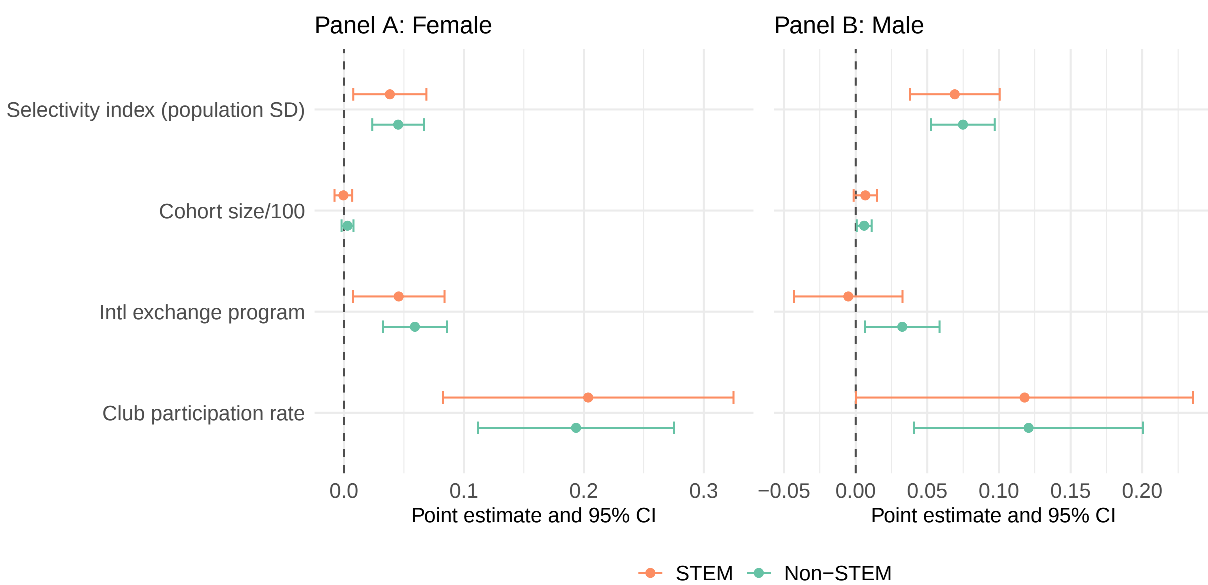
Notes: This figure plots coefficient estimates and the 95% confidence intervals for female and male students where we interact the attributes with an indicator variable for academic abilities, with 50% as the baseline for both groups. Panels A and B interact with an indicator variable for top 50% mathematics abilities students and Panels C and D interact with an indicator variable for top 50% reading abilities students. Standard errors are clustered at the student level. The dotted lines are weighted least squares linear fits of each point, with inverse variance weighting, for each side (below 50% and above 50%) for two groups. We imposed the constraint that the lines pass through the 50% point. The differences in the slopes between the two groups are calculated using the weighted least squares standard errors. Significance levels: * 10%, ** 5%, and *** 1%.

Figure B3: Heterogeneity of Preferences for Other Attributes by Ability



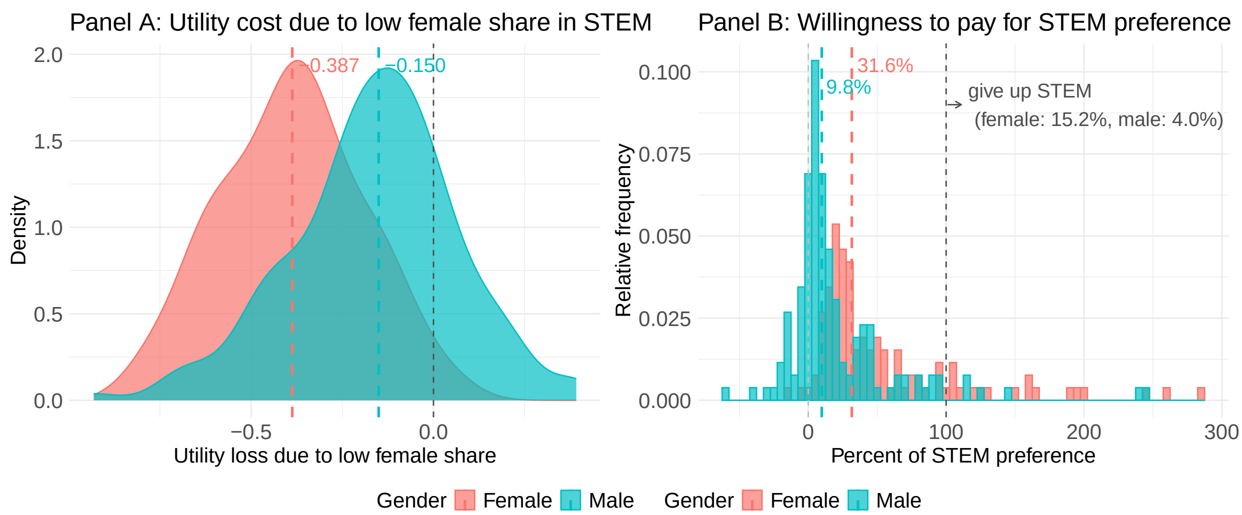
Notes: This figure plots coefficient estimates and the 95% confidence intervals of program attributes for female (Panel A) and male students (Panel B) with different academic abilities. Female student share is included in the estimation but omitted from this figure for brevity. Standard errors are clustered at the student level.

Figure B4: Heterogeneity of Preferences for Other Attributes – STEM vs. Non-STEM



Notes: This figure plots coefficient estimates and the 95% confidence intervals of program attributes for female (Panel A) and male students (Panel B) for STEM and non-STEM programs. Female student share is included in the estimation but omitted from this figure for brevity. Standard errors are clustered at the student level.

Figure B5: Utility Cost and Willingness to Pay



Notes: This figure is restricted to students who prefer STEM. Panel A shows the utility cost due to the low female share in STEM. Panel B shows students' willingness to pay for an increase in the female share, expressed as a fraction of their STEM preference; we exclude the bottom 10% of students by STEM preference, as weak preferences inflate willingness-to-pay estimates.

Table B3: Attribute Values

General attributes	
College name:	AA, AB, AC, AD, AE, AF, AG, AH, AI, AJ, AK, AL, AM, AN, AO, AP, AQ, AR, AS, AT, AU, AV, AW, AX, AY, AZ, BA, BB, BC, BD
Department:	
Non-STEM:	Literature, Law, Business, Economics, Sociology, Foreign Language
STEM:	Physics, Chemistry, Biology, Engineering, Information Technology, Agriculture
Department attributes	
Selectivity index:	55, 57.5, 60, 62.5, 65, 67.5, 70, 72.5
Cohort size:	200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900
Female student share:	5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%
College attributes	
International exchange program:	Yes, Yes, Yes, Yes, No
Club participation rate:	40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%

Notes: This table presents the values each attribute can take in a given program.

Table B4: Preferences for Program Attributes – Reasons as Dependent Variables

Sample:	Female	Male	Female	Male	Female	Male	Female	Male
Outcome:	Fit in		Role model		Studies		Career	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
STEM	-0.037***	0.007	-0.047***	0.011	-0.061***	0.041***	-0.062***	0.021
	(0.012)	(0.013)	(0.013)	(0.013)	(0.015)	(0.015)	(0.016)	(0.017)
Female student share	1.819***	1.099***	0.708***	0.605***	0.239***	0.453***	0.336***	0.407***
	(0.089)	(0.092)	(0.085)	(0.089)	(0.086)	(0.084)	(0.087)	(0.085)
Female student share squared	-1.477***	-1.384***	-0.621***	-0.685***	-0.225***	-0.503***	-0.320***	-0.450***
	(0.094)	(0.093)	(0.084)	(0.089)	(0.084)	(0.086)	(0.086)	(0.086)
Selectivity index (population SD)	-0.014	0.015	0.088***	0.080***	0.084***	0.094***	0.075***	0.080***
	(0.009)	(0.010)	(0.011)	(0.011)	(0.015)	(0.014)	(0.010)	(0.011)
Cohort size/100	0.002	0.005*	0.012***	0.013***	-0.002	0.000	0.004*	0.006**
	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Intl exchange program	0.021	0.020	0.049***	0.040***	0.034***	0.047***	0.053***	0.051***
	(0.014)	(0.013)	(0.014)	(0.012)	(0.013)	(0.013)	(0.014)	(0.015)
Club participation rate	0.282***	0.200***	0.651***	0.443***	0.086**	0.070*	0.165***	0.158***
	(0.042)	(0.043)	(0.054)	(0.050)	(0.039)	(0.041)	(0.040)	(0.043)
Constant	0.504***	0.479***	0.518***	0.490***	0.483***	0.480***	0.494***	0.495***
	(0.007)	(0.007)	(0.007)	(0.008)	(0.007)	(0.008)	(0.007)	(0.008)
Adj. R-squared	0.159	0.100	0.104	0.062	0.032	0.037	0.035	0.029
No. observations	4649	4451	4649	4451	4649	4451	4649	4451
No. students	310	297	310	297	310	297	310	297

Notes: This table presents the same specifications as Table 2 but with indicator variables for the four reasons in place of choice as the outcome variables. Significance levels: * 10%, ** 5%, and *** 1%.

Table B5: Preferences for Program Attributes – Logit

Sample:	Female		Male		All	
Outcome:	Choose the program (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
STEM	-0.078***	-0.078***	0.038**	0.038**	0.037**	0.038**
	(0.018)	(0.018)	(0.018)	(0.017)	(0.017)	(0.017)
Female student share	0.090***	0.819***	-0.036	0.815***	-0.036	0.811***
	(0.022)	(0.097)	(0.024)	(0.097)	(0.024)	(0.094)
Female student share squared		-0.742***		-0.872***		-0.868***
		(0.095)		(0.099)		(0.096)
Selectivity index (population SD)	0.046***	0.046***	0.074***	0.074***	0.074***	0.074***
	(0.011)	(0.011)	(0.012)	(0.012)	(0.012)	(0.012)
Cohort size/100	0.003	0.003	0.006**	0.006**	0.006**	0.006**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Intl exchange program	0.059***	0.060***	0.033**	0.034**	0.032**	0.034**
	(0.014)	(0.014)	(0.013)	(0.013)	(0.013)	(0.013)
Club participation rate	0.193***	0.196***	0.122***	0.130***	0.121***	0.130***
	(0.042)	(0.043)	(0.041)	(0.041)	(0.041)	(0.041)
Female					0.016	0.015
					(0.011)	(0.011)
STEM x Female					-0.116***	-0.116***
					(0.025)	(0.025)
Female student share x Female					0.127***	0.011
					(0.033)	(0.131)
Female student share squared x Female						0.123
						(0.131)
Selectivity index (population SD) x Female					-0.028*	-0.027*
					(0.016)	(0.016)
Cohort size/100 x Female					-0.003	-0.003
					(0.004)	(0.004)
Intl exchange program x Female					0.027	0.026
					(0.019)	(0.019)
Club participation rate x Female					0.074	0.067
					(0.059)	(0.059)
Log Likelihood	-3133.13	-3090.73	-3032.28	-2977.35	-6165.41	-6068.08
No. observations	4649	4649	4451	4451	9100	9100
No. students	310	310	297	297	607	607

Notes: This table presents the logit coefficient estimates in average marginal effects on the program attributes with choice as the dependent variable. Columns 1 and 2 present estimates for female students, columns 3 and 4 present estimates for male students, and columns 5 and 6 present estimates for differences between female and male students. The average marginal effects for constant term are undefined and thus are not shown. Standard errors are clustered at the student level. Significance levels: * 10%, ** 5%, and *** 1%.

C Post-Experimental Questionnaire

Post-Experimental Questionnaire (English translation)

Questionnaire 1/4

Please tell us about yourself and your family.

- Your gender: [Male, Female, Non-binary or Other]
- Your father's academic background: [Below high school, High school, Vocational school, Associate degree, Bachelor's degree, Master's degree or above, I do not know]
- Your mother's academic background: [Below high school, High school, Vocational school, Associate degree, Bachelor's degree, Master's degree or above, I do not know]
- Extra schooling per week: [No extra schooling, one day a week, two days a week, three days a week, four days a week, five days a week or more]

Please recall the exam held on [Month Day]. What was your score in the following subjects?

- Reading: [Integer]
- Mathematics: [Integer]
- English: [Integer]

Please click "→" to proceed.

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Questionnaire 2/4

Do you think **your scores in the exam held on [Month Day] accurately reflect your abilities?** Please answer for each of the subjects below.

	My abilities are lower than the score	My abilities are slightly lower than the score	It reflects my ability accurately	My abilities are slightly higher than the score	My abilities are higher than the score
Reading	○	○	○	○	○
Mathematics	○	○	○	○	○
English	○	○	○	○	○

- Do you consider yourself someone who is **willing to compete with others**, or someone who **avoids competing with others**? [Avoid competing with others, Slightly avoid competing with others, Neither avoid nor willing to compete with others, Slightly willing to compete with others, Willing to compete with others]

- Do you consider yourself someone who is generally **willing to take risks**, or someone who **avoids taking risks**? [Avoid taking risks, Slightly avoid taking risks, Neither avoid nor willing to take risks, Slightly willing to take risks, Willing to take risks]

Please click "→" to proceed.

Page break

Questionnaire 3/4

What do you think is **the average female-student ratio** in the following departments across the colleges in Japan?

	Below 10%	11-20%	21-30%	31-40%	41-50%	51-60%	61-70%	71-80%	81-90%	91% or above
Humanities Departments (Literature, history, philosophy, etc.)	○	○	○	○	○	○	○	○	○	○
Social Sciences Departments (Law, Economics, Sociology, etc.)	○	○	○	○	○	○	○	○	○	○
Sciences and Engineering Departments (Physics, Biology, Mechanical Engineering, Information Technology, etc.)	○	○	○	○	○	○	○	○	○	○
Medicine and Nursing Departments (Medicine, Dentistry, Pharmacy, Nursing, etc.)	○	○	○	○	○	○	○	○	○	○

Please click “→” to proceed.

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Questionnaire 4/4

Please tell us your opinion about this survey.

- Was it easy to follow? [Difficult to follow, Slightly difficult to follow, Neither difficult nor easy to follow, Slightly easy to follow, Easy to follow]
- Which parts did you find it difficult to answer? [Text]
- What do you think is the purpose of this survey? [Text]
- Other comments? (optional) [Text]